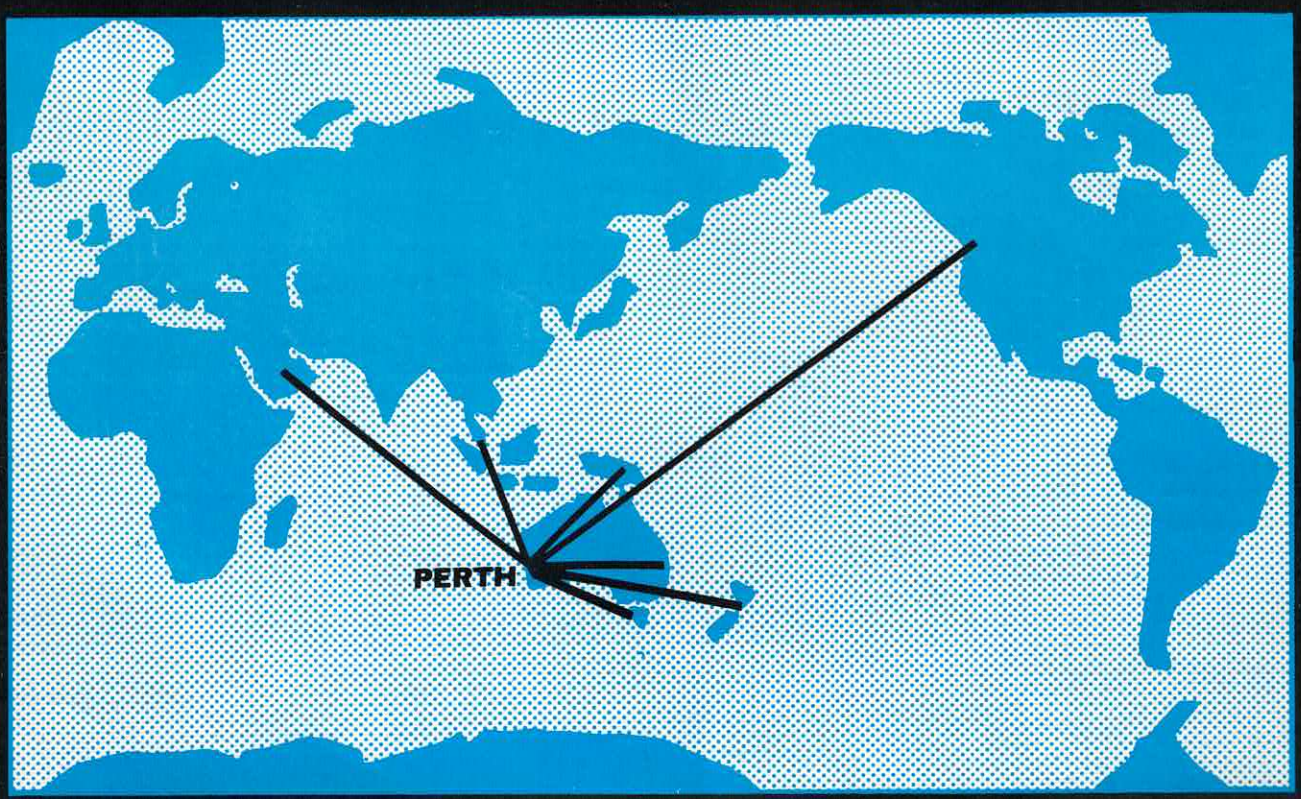
MSX and SPECTRAVIDEO

COMPUTER FORUM

Volume 2 No. 1

Registered by Australia Post — Publication No. WBP1916

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COMPUTER FORUM is a monthly publication published by the following people-

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EDITORIAL

First off I would like to say welcome to all the new subscribers of COMPUTER FORUM.

Secondly I would like to thank all the people that have re-subscribed in showing their faith and support in a new team and new ideas.

As Brod said in his last editorial Vol-1 No-12 the COMPUTER FORUM magazine has been taken over by and I quote "Ross and his team ", I would like to take this opportunity to tell you a little of myself and my team -

I first got interested in computers some 5 years ago, I took a small basic course which started my computer bug, I then bought a SPECTRAVIDEO 318, had that for 2 years, sold that, bought a SPECTRAVIDEO 728 MSX machine and have slowly added various bits of peripheral equipment up to my present stage.

MAX COWIN is our forum treasurer Max also started some years ago in computers and now owns a SPECTRAVIDEO 328 with all the goodies and a SPECTRAVIDEO 738 MSX EXPRESS and has quiet a good knowledge of programming.

TONY DE-IUDICIBUS is our forum secretary Tony has only recently become involved with computers, but his thirst for understanding is admirable and I think that Tony will be a help to us all.

ROBERT STROTHER is our production consultant and legal advisor. Like Tony he has only been into computers for a short time, (but we all started that way). But if we do get in the "poo" for whatever reason I think Bob can pull us out of it.

Please mention COMPUTER FORUM when replying to advertisers.

THE FUTURE

I thought I would tell you a little of the future in the way of ideas we have for COMPUTER FORUM.

We are starting of this issue with a FOR-SALE column for software. Which means any software you feel is good enough to sell we will advertise it, no matter what type of program it is, because of the lack of software in AUSTRALIA for MSX & SV we should all help out the people who are selling software in the market, as well as people making it at home.

I personally think that a program for-sale between $2.00 and $25.00 would sell very well, according to the program type of course. If you sell a program for $4.00 per copy (less handling fees) and sell 50 copys you make about $150.00 which is more than it is worth at your place.

GIVE IT A GO !! (NOTHING VENTURED NOTHING GAINED)

We are also starting up an ADVENTURE column with our very enthusiastic adventurer ALAN CRAWFORD . What we would like from all the Adventurers out there is, what would you like to see in the way of games,hints & maps also if there are any Intrepid Adventurer's out there who have completed games drop us a line and help those of us that are "STUCK".

Also in this issue we will give an incentive to get this magazine really moving, remember the magazine is really run by you we only do the work putting it together.

While I have your full attention I would like to thank (BROD)BRODERICK BROWN for putting together a good magazine and keeping it going for so long, also your support to BROD which I know was quiet an effort from some of you.

BROD is heading off to England shortly and from what he tells us he is going to write a book based on his tutorial that he made in the Magazine so I'm sure you all wish him the best of luck.

So from ME and my TEAM to YOU and BROD THANKYOU.
And I hope we recieve the same support that you gave BROD.

READERS ADS

Use these pages to sell your surplus equipment, etc. Photocopy this form and insert up to 20 words. Put it in an envelope with $5 and send to COMPUTER FORUM by the 10th of each month.

..... : ..... : ..... : .....

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****
******** PLEASE NOTE THAT WE HAVE CHANGED OUR ADDRESS ********
****

INCENTIVE FOR FREE ISSUES OR PRESCRIPTIONS

One other thing we have introduced into COMPUTER FORUM now is an incentive scheme which we think could be a benefit to us all, not only will it boost up sales giving us power in numbers, as well as varied ideas in programs, but we can improve our magazine to the point where we can match the best computer magazines around.

THE IDEA

All you have to do is find someone who has an MSX or SPECTRAVIDEO computer get them to subscribe to COMPUTER FORUM and include your name, and you will recieve 1 FREE issue of COMPUTER FORUM for every 12 months subscription that you can get people to sign up for. If you get 12 people your subscription for the next 12 months is FREE.

********** SOFT-ADDS **********

FTYPE2. (SA1) This is a typing program which may help speed up your typing. MSX **Price $4.00**	FIND-ME. (SA2) A game where you play against the computer to find his men before he finds yours. MSX **Price $3.00**
LOTTO. (SA3) A game which chooses your numbers and puts them out to the printer or plotter or plays your game. A good program. MSX **Price $3.00**	SORT. (SA4) A small utility program that sorts out mames or numbers and puts them to a printer or plotter if required. MSX **Price 2.00**

MINIMUM CHARGE $12.00 : PLEASE QUOTE (SA) NUMBER WHEN ORDERING.
As you can see this is only a start up of the SOFT-ADDS column, but I'm sure with your help it will expand.
Send in those programs and make a little extra money or maybe a lot.

PROGRAMMING HELP

I am writing a short word processor because Just Write Jnr doesn't take full advantage of the printer's capabilies.
The problem is that the computer keeps doing garbage collection (what you were talking about in the last edition) and I'd like to know if I could perform the same operation within the program so ther is no interuption to the typing.
My program allows you to enter control codes for the printer.

Peter A Beil Kumbia,Queensland

Peter, without viewing your program, the only suggestion we can make is to clear more string space using the CLEAR command and/or limit input to one screen full before printing it out.

If there are any of our readers that may be able to solve Peters problem please drop us a line.

SOFTWARE ADVERTISING

RULES

We thought of various ways that we could do the software advertising, as to please everbody, we then came to the following agreement-

When advertising your own home-made software you must observe the following rules -

1.. It must be your own Home-made software

2.. You can not re-sell sotware from software houses.

3.. The program or programs can not be any shorter than 30 lines if renumbered by increments of 1.

4.. Computer Forum reserve the right to refuse software advertising.

5.. Computer Forum have the right to ammend this agreement.

6.. Computer Forum will hold all the Authors money until it reaches a minimum of $50.00 then it will be sent to the Author.
(in certain cases this may not apply.)

COSTS

Computer Forum will handle all software in-coming and out-going.

The basic fee for advertising your software which includes a small review is $5.00.

Handling fees for your software works out to $4.00 or 25% of its selling price.

(Handling fees include Tape or disk,copying time & verification, postage to the buyer and postage of money to Author)

Example -
5 games = a selling price of $16.00, - $4.00 or 25% handling fee
you recieve $12.00 x 30 people = $360.00

Try to keep your prices with-in reason, and people will buy it.

Address the form enclosed to - COMPUTER FORUM
SOFT-ADS
PO BOX 87
MT HAWTHORN 6016
WESTERN AUSTRALIA

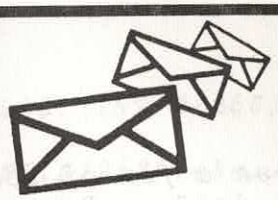
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MAILBOX



AUSTRALIAN ADVENTURE MASTERS CLUB *for adventurers.....*
If you are involved with Adventure Games and are stuck somewhere you may be interested in contacting us.
Membership is FREE but you must be prepared to help others who are in the same boat. For more information please ring IVAN on -(09) 2794460 or write to- 197 ANZAC TERRACE,BASSENDEAN 6054,W.A.

Here is a letter from G.ARCHER of Brisbane.
He wishes to know if any of our readers have a screen dump program for an MSX computer (Pioneer PX7) and a printer plotter (Toshiba HX-P570). If so we would like to hear from you.

********** ***PEN PALS*** **********

This is an idea sent into us by one of our readers.
So if there are any pen holders still left or nimble fingered typists who would like to get together and form a Pen-Pals club, Send us a stamped self-addressed envelope with your name and address and we will send you a list of all the people you can write/type to in the PEN-PALS CLUB.

We have had 3 replies to the TONY JAMES question in volume 1 No 12, "How do I turn my drive off faster", and to that question I would like to thank Tony Jaffrey & Wes Nicholson for their effort, also I would like to thank Mr A. Nonymous for his suggestion that follows !!!

Suggestion 1..

```
10 REM Using an architects square carefully formulate the exact
20 REM centre of your disk drive and mark it clearly.
30 REM (Now comes the tricky bit)
40 REM To stop your disk drive VERY fast use a 1.2kg hammer,and
50 REM bash the marked area until the drive stops.
60 REM (This program will also work on MSX drives)
70 REM !! CAUTION !! If used incorrectly file damage could accure
```

Suggestion 2..

```
10 CLEAR 1000
20 FIELD #0,30 AS Z$
30 A$=DSKI$(1,0,12)
40 MID$(A$,20,1)=CHR$(50)
50 MID$(A$,21,1)=CHR$(0)
60 LSET Z$=LEFT$(A$,30)
70 DSKO$ 1,0,12
```

I think suggestion 2 would probabally be the better choice.

Here is an offer from the SVI & MSX users club in Brisbane.
They wish to offer to any member of any SVI or MSX user club a program that works on the SV328 or expanded SV318. It is a conversion program that works for most MSX programs written in basic. It is a tape-based system only, but may be loaded initially from disk to tape. There may be some diffuculty in saving a program accurately to a SV903 data cassette. However programs saved on the MSX cassette will load successfully. All enquiries to -THE SECRETARY,BRISBANE SVI & MSX USERS GROUP,QUEENSLAND,4114. Phone-(07) 2085951.

MSX GAMES BOOK

I suppose a few of you have a had problems with the MSX games book, well Mr Jeffrey Dean of Victoria did so he decided to do something about it, he wrote a letter to the publishers of the book MELBOURNE HOUSE and they sent back a reply saying they apologise for any inconvenience.
So to all the people having problems try these fixups-

Page 8

The heading CHEXSUM PROGRAM is incorrectly placed on this page. It should appear directly after "Finder Program"(before the paragraph "The value of SMEM=32767...etc").

Page 9

A line was left out of the Chexsum Table.
Please insert in the table - 62180=4133

Page 10

Refer end of item 5 and delete "Entering a P will cause output to go to a printer, S to go to a screen.
Insert in it's place -
If a printout is desired you must insert your own printing routine before line 62015. This routine is the same as lines 62020 to 62230 except that all PRINT statements are replaced by LPRINT statements.

Page 20

In the program ASTRAL ATTACK,change line 10000 to read-

```
DATA 1,1,1,65,65,199,206,206,254,206,254,207,199,198,199,67,128,
     128,128,130,130,227,115,115,127,115,127,243,227,99,227,194
```

Page 21

Because of the above change in line 10000, change the Chexsum table to-
Line 10000 = 10169
Total = 130821

Page 111

In the program CHOPPER LANDING change line 300 to read -

```
300 K=VPEEK (6912):IF K 4 OR K  191 THEN VPOKE 6912,4:RETURN
```

Page 161

LASER BATTLE
Some implementations of MSX BASIC do not support ¥ for integer division. If ¥ signs are keyed in but dissapear when listed, the following changes must be made-

Page 164

Line 165 POKE 59992!, ISP MOD 256:POKE 59993!,INT(ISP/256):I3=6720 + PG:POKE 59994!,I3 MOD 256:POKE59995!,INT(13/256):etc..as listed

****
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****

FIX-UPS CONTINUED

Line 193 Type as listed to I MOD 256:POKE 59993!,INT(I/256)

Line 200 POKE 59994!,J MOD 256:POKE 59995!,INT(J/256):etc..as listed

Page 166 Line 282 Type as listed to I MOD 256:POKE 59991!,INT(I/256)

Line 290 POKE 59994!,J MOD 256:POKE 59995!,INT(J/256):D= USR2 (D)

Page 169 The titles of the grids are the wrong way around please reverse titles of LASER and MOTHER SHIP.

Page 203 In the program MINOTAUR MASTERMIND change line 654 to read- GOSUB 900:K2 etc..as listed

I know it looks like a lot of mistakes but when you take into account the size of the book MELBOURNE HOUSE did do a good job.

MSX and SVI SPECIAL



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Aerobat: Cessna 152 flight simulator. Written in machine code and was a finalist in the New Zealand Software awards for SVI. Was $22.50, now $12.

Chess: Two level chess game with a tutorial for your SVI. Was $22.50, now $12.

Memory Expansion: Use both banks of your SVI or MSX and get a total of 60K true user ram. Was $22.50, now $10.

Various other bargains available. All prices include postage. This is as cheap as we can go folks, so get it while you can.

HARDWARE FOR SALE

1 x SV-328 Keyboard $50.00	SONY HIT BIT(64K),DISK DRIVE
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########################## HIGH GAME SCORE ##########################

In order to make this a success we need some active people to send in their high scores of any game they have tackled, then we can make a list second to none (or at least close).
What we need from you is how you went with those nasty creepy grotty not so nice guys, and how you got up to LEVEL 76,456,596 without stretching the truth just a little.
So lets keep it honest and we may have a true "HIGH GAME SCORE" list that everybody can use to test their ability...

Send your name,name of game & score to- COMPUTER FORUM
HIGH GAME SCORE
PO BOX 87
MT HAWTHORN 6016
WESTERN AUSTRALIA

USER GROUPS

Adelaide area should contact
JOE ROWLEY
255 Hogarth Road, Elizabeth Grove
South Australia 5112. Phone 255 6738

SYDNEY USERS

We have been advised of a group in Sydney for Spectravideo owners– please contact Malcolm Samuels on 888 7664 after hours.

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25 Primrose Street, Woodridge, Queensland 4114

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Software NEWS & REVIEWS



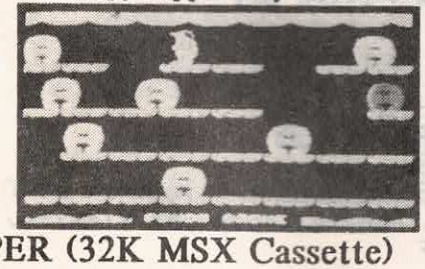
ZOOT (64K MSX Cassette)

Zoot has lost his marbles down a drain. He must go and retrive them, but beware, Zoot! There are strange things that live in drains. They're called Mankins and there are nine of the horrible things wandering around down there. Zoot must either punch or trap them, avoid them and out-manoeuvre them to get his marbles back again.

Unusual concept, good graphics - no music.

Publisher: Bug-Byte
Price: $12.95

Review copy supplied by LPG Software

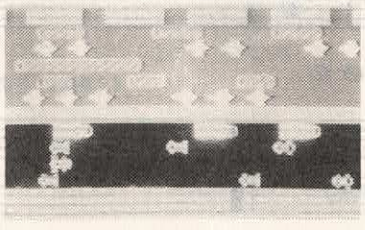


HOPPER (32K MSX Cassette)

What can we say about this one? It is the old classic which goes by the names Hoppy, Frogger, Toader etc. where you must get the frog across the road and across the river. This one has good graphics and smooth action for the price.

Publisher: Aackosoft
Price: $14.95

Review copy supplied by LPG Software



SASA (SV318/328 Cassette)

Sasa is the latest highly trained robot and his purpose is to fight astro-monsters. To do this, he must first complete three energy gathering stages. He begins by attempting to blast his way into an arsenal. Once there, he must destroy the power generators which will expel energy bottles into the sea. Sasa must destroy the octopi in the sea and collect these energy bottles and proceed to do battle with the fierce astro-monsters. Go for it, Sasa!

1 or 2 players, good graphics, music and effects, 4 screens, 4 levels - good concept.

Requires extra memory for use with SV318

Publisher: Spectravideo
Price: $19.00

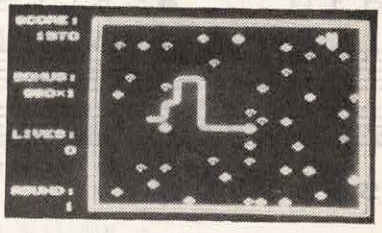
SNAKE IT (64K MSX Cassette)

Seen it before? Not this one! Sure it is based on the old original game of moving the snake around and eating clover and flowers so that the snake grows, but this one has more than five screens - we think. We haven't got past the fifth screen. On screen 4 you must also move blocks around to get at the clover and flowers.

Smooth action and cleverly written.

Publisher: Aackosoft
Price: $14.95

Review copy supplied by LPG Software



HINTS & TIPS

These tips are for the owners of the YAMAHA CX5M music computer and may help with their YRM-101 composer software. I use this method when portamento (sliding pitch) is required However, it only works when sliding up, not down.

Firstly prepare a data-tape of user voices (YRM-102 software) by modifying the chosen voice as shown in the L.F.O. section (top right of fig.A.).
Use internal voice No 7 for the TEST SCORE.
Now load the test music score into YRM-101 Composer software exactly as illustrated.
Follow that with your data-tape of voice information and play the result.-- Hey presto, PORTAMENTO.

Incidentally, to print the score in large format if using a YAMAHA PN-101 printer (and most others I imagine) try the following program while in basic. Then leave the printer switched on while changing over to the composer cartridge for the printout.

```
20 LPRINT"NORMAL"CHR$(14)"ENLARGED"CHR$(13)CHR$(10)
30 END
```

Courtesy of Russell Zimmer

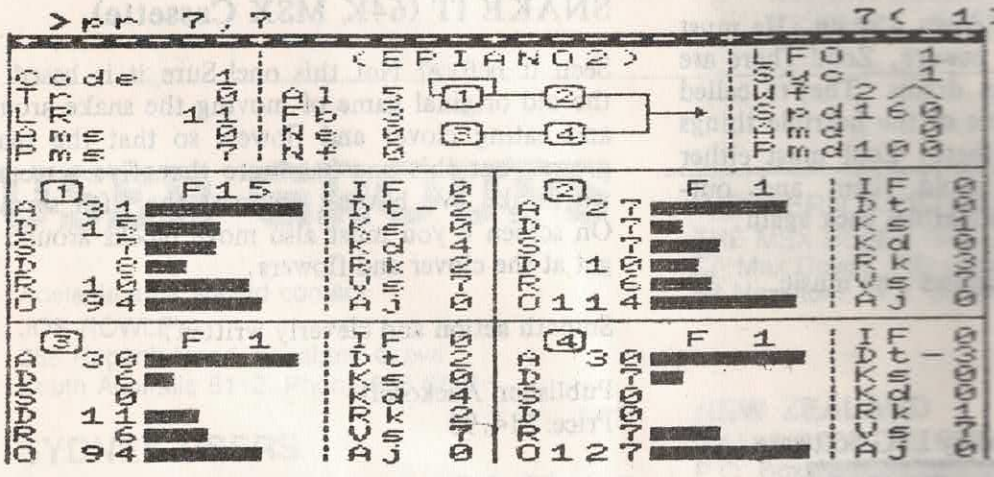
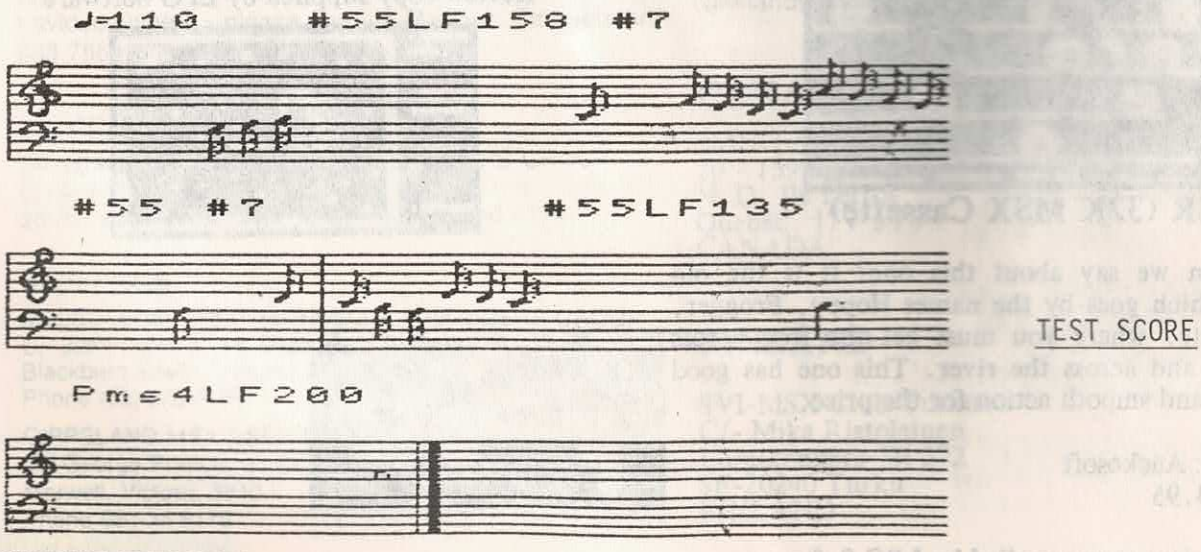


FIG. A



TEST SCORE

COMPUTER FORUM
Magazine
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BASIC TUTORIAL CONCLUSION

IMPROVE YOUR PROGRAMMING SKILLS

In this, the final part of the tutorial, we will look at some of the different uses for the commands which have been covered. Although every endeavour has been made to include worked examples wherever possible, this can lead the reader to assume that the example given is the only way in which that command can be used.

SPRITES

Firstly, let us look at the different ways in which a Sprite can be built. In all of the following examples you will need to add line 10 to set the screen - MSX use SCREEN 2,2 and SV use SCREEN 1,2

The simplest sprite to build is the 8 x 8 block sprite using

Example 1:
```
20 SPRITE$(1)=STRING$(8,255)
```

This is a lot more simple than using what is known as the 'sprite building loop' which would look like this

Example 2:
```
20 FOR I=1 TO 8
30 READ A
40 S$=S$+CHR$(A)
50 NEXT I
60 SPRITE$(1)=S$
70 DATA 255,255,255,255,255,255,255,255
```

This example uses a FOR/NEXT loop to read eight data values into the variable S$ which is then assigned to sprite number one in line 60.

The same thing could be done using binary data values -

Example 3:
```
20 FOR I=1 TO 8
30 READ A$
40 S$=S$+CHR$(VAL("&B"+A$))
50 NEXT I
60 SPRITE$(1)=S$
```

```
70 DATA 11111111
71 DATA 11111111
72 DATA 11111111
73 DATA 11111111
74 DATA 11111111
75 DATA 11111111
76 DATA 11111111
77 DATA 11111111
```

We could also use hexadecimal values to generate the sprite -

Example 4:

```
20 FOR I=1 TO 8
30 READ A$
40 S$=S$+CHR$(VAL("&H"+A$))
50 NEXT I
60 SPRITE$(1)=S$
70 DATA FF,FF,FF,FF,FF,FF,FF,FF
```

Another method of using hexadecimal values is -

Example 5:

```
20 V$="FFFFFFFFFFFFFFFF"
30 FOR I=1 TO 16 STEP 2
40 S$=S$+CHR$(VAL(MID$(V$,I,2)))
50 NEXT I
60 SPRITE$(1)=S$
```

Although the latter method looks cumbersome it has the advantage of being memory efficient when providing data for a number of sprites.

Let us now look at the different ways of building several sprites using the methods just described. Firstly we will build six 8 x 8 sprites.

Example 6:

```
20 FOR S=1TO6
30 SPRITE$(S)=STRING$(8,255)
40 NEXT S
```

In this example we have used the same method as example 1 to build the sprite and enclosed it within a FOR/NEXT loop to obtain the six sprites.

Example 7:

```
20 FOR S=1 TO 6 : S$="" : RESTORE 90
30 FOR I=1 TO 8
40 READ A
50 S$=S$+CHR$(A)
60 NEXT I
70 SPRITE$(S)=S$
80 NEXT S
90 DATA 255,255,255,255,255,255,255,255
```

In line 20 we cancel the old value of S$ with S$="" and to save having six lines of data statements which are all the same, we use RESTORE 90

to keep re-reading the same data.

We do the same in the next example except we use another RESTORE in line 80 to save having a total of 48 data statements -

Example 8:

```
20 FOR S=1 TO 6 : S$="" : RESTORE 100
30 FOR I=1 TO 8
40 READ A$
50 S$=S$+CHR$(VAL("&B"+A$))
60 RESTORE 100
70 NEXT I
80 SPRITE$(S)=S$
90 NEXT S
100 DATA 11111111
```

This next example also uses the same piece of data 48 times -

Example 9:

```
20 FOR S=1 TO 6 : S$="" : RESTORE 100
30 FOR I=1 TO 8
40 READ A$
50 S$=S$+CHR$(VAL("&H"+A$))
60 RESTORE 100
70 NEXT I
80 SPRITE$(S)=S$
90 NEXT S
100 DATA FF
```

The next example uses the sprite building technique of example 5 but because the value of V$ never changes, we do not need to use RESTORE to allow re-reading of the same information.

Example 10:

```
20 V$="FFFFFFFFFFFFFFFF"
30 FOR S=1 TO 6 : S$=""
40 FOR I=1 TO 16 STEP 2
50 S$=S$+CHR$(VAL(MID$(V$,I,2)))
60 NEXT I
70 SPRITE$(S)=S$
80 NEXT S
```

All of these examples have assumed that we want each of the sprites to be the same. Let us now look at methods of constructing three different sprites with the different methods already covered.

Example 11:

```
20 SPRITE$(1)=STRING$(8,255)
30 SPRITE$(2)=STRING$(4,192)+STRING$(4,7)
40 SPRITE$(3)=CHR$(255)+STRING$(6,129)+CHR$(255)
```

Example 12:

```
20 FOR S=1 TO 3 : S$=""
30 FOR I=1 TO 8
40 READ A
50 S$=S$+CHR$(A)
60 NEXT I
70 SPRITE$(S)=S$
80 NEXT S
```

```
90 DATA 255,255,255,255,255,255,255,255
100 DATA 192,192,192,192,7,7,7,7
110 DATA 255,129,129,129,129,129,129,255
```

Example 13:

```
20 READ A$ : A=VAL("&B"+A$) : FOR I=1 TO 8
30 S$=S$+CHR$(A) : NEXT I : SPRITE$(1=S$
40 READ A$ : A=VAL("&B"+A$)
50 READ B$ : B=VAL("&B"+B$)
60 SPRITE$(2)=STRING$(4,A)+STRING$(4,B)
70 READ A$ : A=VAL("&B"+A$)
80 READ B$ : B=VAL("&B"+B$)
90 SPRITE$(3)=CHR$(A)+STRING$(6,B)+CHR$(A)
100 DATA 11111111
110 DATA 11000000,00000111
120 DATA 11111111,10000001
```

Example 14:

```
20 READ A$ : SPRITE$(1)=STRING$(8,VAL("&H"+A$))
30 READ A$ : A=VAL("&H"+A$)
40 READ B$ : B=VAL("&H"+B$)
50 SPRITE$(2)=STRING$(4,A)+STRING$(4,B)
60 READ A$ : A=VAL("&H"+A$)
70 READ B$ : B=VAL("&H"+B$)
80 SPRITE$(3)=CHR$(A)+STRING$(6,B)+CHR$(A)
100 DATA FF,C0,07,FF,81
```

Example 15:

```
20 V$(1)="FFFFFFFFFFFFFFFF"
30 V$(2)="C0C0C0C007070707"
40 V$(3)="FF818181818181FF"
50 FOR S=1 TO 3 : S$=""
60 FOR I=1 TO 16 STEP 2
70 S$=S$+CHR$(VAL(MID$(V$(S),I,2)))
80 NEXT I
90 SPRITE$(S)=S$
100 NEXT S
```

USING RELATIONAL OPERATORS

Relational operators were introduced in part two. These were -

= equal
< less than symbol
> greater than symbol
<= less than or equal
>= greater than or equal
<> not equal

These can be used in different ways. The most obvious way is not always the best. For example -

```
10 INPUT A
20 B=20
30 IF A>B THEN PRINT "TOO BIG":GOTO 10
40 PRINT B
```

could have been written

```
10 INPUT A
20 B=20
30 IF A<=B THEN PRINT B
40 PRINT "TOO BIG":GOTO 10
```

The following examples all perform the same function -

```
IF A=B THEN PRINT "EQUAL"
IF B=A THEN PRINT "EQUAL"
IF A<B+1 AND A>B-1 THEN PRINT "EQUAL"
```

and these perform the same as each other -

```
IF A>B THEN PRINT "NOT EQUAL"
IF A<B THEN PRINT "NOT EQUAL"
IF A<>B THEN PRINT "NOT EQUAL"
```

to check if a number is greater than 100

```
IF N>100 THEN do something
IF N>=101 THEN do something
IF N<101 THEN don't do something
```

to check if a number is less than 100

```
IF N<100 THEN do something
IF N<=99 THEN do something
IF N>99 THEN don't do something
```

The RETURN command

Although this command has already been covered in part two, it was not explained that a line number can follow the command. If the line number is left out, then program flow will return to the command following the GOSUB which called the subroutine.

Example:
```
10 GOSUB 100
20 PRINT "I HAVE RETURNED"
30 END
100 PRINT "I AM AT LINE 100"
110 RETURN
```

Sometimes we will want to use a 'conditional return'. That is, if the

subroutine checks for something and it is true then we might want to return to a different place in the program than if it were false.

```
Example:  10 INPUT"NAME";N$
          20 GOSUB 100
          30 PRINT "NAME"
          40 PRINT "BYE FOR NOW"
          50 END
          100 IF N$<>"SUSAN" THEN RETURN 40
          110 RETURN
```

This little program asks for a name and then goes to a checking routine at line 100. If the name is not SUSAN then it returns to line 40 and ends the program. If the name is SUSAN then it returns to line 30 and prints the name. This can be useful for having passwords at different levels of your program.

SPEEDING UP PROGRAMS

There are several methods and commands for speeding up the performance of programs.

1. If only whole numbers are used in a program, use DEFINT at the start of a program. This will define as integer, or whole numbers, the calculations made within a program. The computer does not have to waste time working out all of the decimal values within those calculations.

2. When a program is run, and the computer is searching for line numbers given in GOTO and GOSUB commands, it starts looking for them from the start of the program. To speed up this search, place regularly used subroutines at the start of the program and the less time dependant routines, such as instructions and title screens, at the end of the program.

3. Wherever possible use multi-statement program lines. This also speeds up loading from tape. Each program line has what is known as a five byte overhead. The line number uses two bytes, then pointer to the next line number takes another two bytes and the end of line indicator takes one byte.

4. Try to anticipate the normal response to IF/THEN commands and use them in the sequence which would allow action to be taken with the least number of checks. For example, if the normal response is YES then don't check for NO first -

```
Wrong     10 INPUT "ENTER YES OR NO";A$
          20 IF A$="NO" THEN END
          30 IF A$="YES" THEN 10
```

Right

```
10 INPUT "ENTER YES OR NO";A$
20 IF A$="YES" THEN 10
30 IF A$="NO" THEN END
```

MEMORY CONSERVATION

The least amount of space used by a program will also speed up operation and loading. The more efficient a program then the less memory required to run it.

1. Use the same variables over again once you have finished with them. This is an example of sloppy programming (the information in brackets is the number of bytes consumed by each line) -

```
10 FOR I=1 TO 100          (15 bytes)
20 PRINT I                 (8 bytes)
30 NEXT I                  (8 bytes)
40 FOR J=1 TO 100          (15 bytes)
50 PRINT J                 (8 bytes)
60 NEXT J                  (8 bytes)
70 FOR K=1 TO 100          (15 bytes)
80 PRINT K                 (8 bytes)
90 NEXT K                  (8 bytes)
```

This program consumes 93 bytes. It could have been written -

```
10 FORI=1TO100:PRINTI:NEXTI        (18 bytes)
```

This is just the amount of memory used to store the program. Another saving is made by the fact that each time a FOR/NEXT loop is used, the information concerning each loop stays in the Stack Area of memory UNTIL THAT VARIABLE IS USED IN ANOTHER LOOP.

2. If you must jump out of a FOR/NEXT loop, don't use this -

```
10 FOR I=1 TO 100
20 C=C+1
30 IF C=50 THEN 50
40 NEXT I
50 PRINT C
```

Each time you jump out of a loop without properly terminating it, you will leave 25 bytes in the Stack Area which is never recovered. Better to use this version -

```
10 FOR I=1 TO 100
20 C=C+1
30 IF C=50 THEN I=100
40 NEXT I
50 PRINT C
```

We have changed line 30 to con the computer into thinking it has looped 100 times and therefore exit the loop in the normal way.

USING DISKS TO SAVE MEMORY

If you are one of those fortunate enough to have disk drives, then you can write programs which are larger than the available memory space - with a little thought.

If you were to write a data base program, then the input section could be one program, the search section another and so on. All of these sections could be loaded and executed from a MENU program. An overview of such a design would probably look like this -

MENU	List functions Input Search Sort Alterations Print Select function Load program to perform that function
INPUT	This program accepts input from the keyboard and stores it on the disk. The last line would load the MENU
SEARCH	This program would search for a record on the disk and display it if found. The last line loads the MENU
SORT	This program sorts all of the records and then loads the MENU
ALTERATIONS	This program uses part of the INPUT program to get the updated information. It then uses the SEARCH program to find the old information on the disk and make the alterations. It then loads the MENU
PRINT	This program prints the information contained on the disk and loads the MENU

This method of programming is called MODULAR CONSTRUCTION with smaller programs, or modules, performing a specific task. The modules are all tied together with the menu program.

PROGRAM PLANNING

When you write your own programs, a little bit of preliminary planning work can save problems later. For example, if you want to write a

program which will be used for storing names and addresses, then the following steps might be considered.

1. List what it is that the program should do

   Entry of names and addresses
   Display entries
   Allow update of entries - Additions, alterations and deletions
   List all entries on a printer
   Search for a given entry
   Sort into alphabetical order
   Load from tape/disk
   Save to tape/disk

2. Detail what each of the above functions should do in more detail
   For example - Entry of names and addresses
   Due to the need for sorting on surnames, then we cannot have the first name or initials entered in the same field as the surname unless it is entered surname first (eg DAVIS S.G.)
   How will we handle bad information - will we clear the screen and start again or just blank out the current input field and request the correct input?

3. When we have done all of this, then write the program in modular form. That is, write a program which will allow input of names and addresses and completely debug that. When that is working properly, write the next module which might be the display function and debug that. Now renumber this module using RENUM and MERGE it with the first module and so on until the whole program comes together.

4. Organise some test data and try the program to see if it does what you originally intended.

It is far easier to write and debug small modules than a whole program which does not seem to work. Keep your original modules separate so that alterations can be made and tested if needed at a later stage - you may want to add telephone numbers or something. This is more easily done with small modules than the whole program.

* * * * *

This concludes the Basic Tutorial. However, with computers, the learning has just begun.

The contents of this series of tutorials is the property of -

Broderick J Brown

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ANSWERS to the questions in Part 12

1. (a) MSX uses KEY OFF
   (b) SV318/328 use SCREEN 0,0

2. OUT OF STRING SPACE

3. Using PUT in this instance is an SV318/328 graphics command.

4. BASE is an MSX command.

5. SV318/328 use only screen numbers 0 to 2.

QUESTIONS

1. Which of these is wrong?

   (a) A=10:B=20:A<>B
   (b) A=10:B=20:A<B
   (c) A=10:B=20:A>=B

2. Debug this program -

```
10 A=10
20 GOSUB 100
30 PRINT A
40 END
50 A=A*2
60 RETURN
```

3. What is wrong with this?

```
10 PRINT "10"
20 GOSUB 100
30 PRINT "30"
100 PRINT "100"
110 RETURN
```

4. What is wrong with this statement?

```
LINE(20,40)-(100,40)
```

ANSWERS: 1 (c),
2 line 20 should be GOSUB 50,
3 there is no END command before the subroutine
4 letter O used instead of zero

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HACKERS CORNER

Hi, after a long spell here I am back again writing HACKER'S CORNER. If anyone in W.A. knows of some 5 1/4" disk drives for less than $200.00 please contact me.
Come on hackers, I need input from you in either suggestion, criticism or praise form. As a rule, machine code games sell fairly well and if you have written a game, utility or other in machine code then please get it out of your computer and onto the shelves of a computer store.

CP/M CORNER

As requested by some readers I now start the first of my many articles on CP/M which is primarily based on the SV-328 computer.
To get into Machine Code programming you should have on a disk: Macro 80, Link 80, Wordstar or ED, DASMZLG.
To summarise, Macro 80 is a very good macro assembler which will assemble both 8080 and Z80 code, Link 80 is necessary to have because after a file is assembled it must be converted from a .REL (relative) file to a .COM (command) file.
Next a text editor is needed, use your own judgement getting this program as simple line editors like ED are not really worth anything at all in the way of functions but Wordstar is very good if you have 80 columns but dont forget it if you havn't.
It seems that a 40 column text editor is in great need of being written.
To really get into the hang of things, there is a public domain dis-assembler around which is more, much more than a straight out dis-assembler. Why you ask, well because the program in question DASMZLG, is a symbolic dis-assembler which quiet simply means that say for example your friend gave you a nice M/C utility program and you wanted to see how it works, with DASMZLGyou can create an .ASM file from that program which you are dis-assembling. Interested ?.
If so then look at next months CP/M section where I will be describing DESMZLG in more detail.

NEED INPUT

As No5 on the the film -SHORT CIRCUIT- says "I NEED INPUT", well this is exactly the way I feel, as writing Hacker's Corner is involved at times and some feedback from my articles would make a very big difference. But what I must stress is that I am not about to include a section or tutorial on teaching Machine Code programming.
This may sound very negative but the point being, it is one aspect of computers that really only trial and error can make you learn the so called "tricks of the trade". I will of course simplify my articles to a degree where normal and sane people (non-hackers can actually read my articles.
Programs like my DOS I designed could with the proper feedback (thanks to those 2 people who wrote in) been the new medium to transport programs between the SVI and MSX computers. I was thinking about a design to add-on a 3 1/2" disk drive to the SV-318,328 computers and coupled up with the DOS would ensure that MSX programs could be read on the SV and visa versa. Well they are in the pipeline anyway, but for now I really do need input from you.

THIS PROGRAM IS MADE TO SUIT
THE SV-328 OR 318

```
10 REM------------------------------
20 REM---
30 REM---
40 REM---
50 REM------------------------------
60 CLEAR500
70 GOSUB1620
80 COLOR15,1,10:SCREEN1,2
100 REM--------MAIN VARIABLES--------
120 P1%=50:P2%=220:P3%=45:P4%=45:P5%=40
:P6%=60:P7%=88:P8%=80:P9%=83:P1=15:P2=1
0:P3=9:TE=1:X=4:Y=23:MN=4:P=2:Z=1:NAM=1
:A=103:DRW=15:RF=148:SC=0:RY=1:FJ=1
130 GOSUB1030
140 GOSUB1290
160 REM------JOY STICK SEQUENCE------
180 IFRY=1THENSOUND8,0ELSE190
190 T=(STICK(1)ORSTICK(0))
200 IFX<1THEN1950ELSEIFX>235THENX=235
210 IFT=1THEN220ELSE230
220 IFP=2ORP=6THENP=1ELSEIFP=4ORP=3THEN
P=5ELSE230
230 IFT=2ANDTE=1THENX=X+4:P=1:IFTE=2THE
NX=X-4:P=1
240 IFT=3ANDTE=1THENX=X+4:P=2:IFTE=2THE
NX=X-4:P=2
250 IFT=4ANDTE=1THENX=X+4:P=6:IFTE=2THE
NX=X-4:P=3
260 IFT=5THEN270ELSE280
270 IFP=2ORP=1THENP=6ELSEIFP=4ORP=5THEN
P=3ELSE280
280 IFT=6ANDTE=1THENX=X-4:P=3:IFTE=2THE
NX=X+4:P=6
290 IFT=7ANDTE=1THENX=X-4:P=4:IFTE=2THE
NX=X+4:P=4
300 IFT=8ANDTE=1THENX=X-4:P=5:IFTE=2THE
NX=X+4:P=4
320 REM-------POSITION SPRITES-------
340 PUTSPRITE1,(X,Y),15,P
350 PUTSPRITE2,(180,175),P1,P
360 PUTSPRITE3,(200,175),P2,P
370 PUTSPRITE4,(220,175),P3,P
380 PUTSPRITE5,(P1%,23),7,7
390 PUTSPRITE6,(P2%,34),15,8
400 PUTSPRITE7,(25,51),10,9
410 PUTSPRITE8,(47,P3%),5,10
420 PUTSPRITE9,(110,P4%),15,11
430 PUTSPRITE10,(210,P5%),10,12
440 PUTSPRITE11,(P6%,73),2,8
450 PUTSPRITE12,(17,A),3,13
460 PUTSPRITE13,(44,P7%),8,14
470 PUTSPRITE14,(110,P8%),15,15
480 PUTSPRITE15,(168,P9%),10,16
490 PUTSPRITE16,(228,158),15,17
500 PUTSPRITE17,(207,147),15,17
510 PUTSPRITE18,(187,158),15,17
520 PUTSPRITE19,(166,147),15,17
530 PUTSPRITE20,(146,158),15,17
540 PUTSPRITE21,(124,147),15,17
550 PUTSPRITE22,(104,158),15,17
560 PUTSPRITE23,(30,125),10,18
580 REM-----MOVEMENT FOR SPRITES-----
```

ROAD RUNNER

by Malcolm Brand

```
600 P1%=P1%+4:IFP1%>220THENP1%=50
610 P2%=P2%-3:IFP2%<50THENP2%=220
620 P3%=P3%+2.5:IFP3%>60THENP3%=45:RY=2
630 P4%=P4%+2:IFP4%>58THENP4%=45:RY=2
640 P5%=P5%+1.5:IFP5%>60THENP5%=40:RY=2
650 P6%=P6%-3:IFP6%<33THENP6%=210
660 P7%=P7%+1:IFP7%>109THENP7%=88:RY=2
670 P8%=P8%+2:IFP8%>109THENP8%=80:RY=2
680 P9%=P9%+2.5:IFP9%>109THENP9%=79:RY=
2
690 IFRY=2THEN700ELSE710
700 SOUND0,109:SOUND1,61:SOUND8,12:SOUN
D7,8:SOUND11,61:RY=1
710 SPRITEON:ONSPRITEGOSUB760
720 GOSUB1840:GOTO160
740 REM------COLLISION SEQUENCE------
760 SPRITEOFF:MN=MN-1:IFMN=0THEN920ELSE
770
770 SOUND8,0:SOUND1,15:SOUND6,4:SOUND7,
54:SOUND8,16:SOUND12,12:SOUND13,0:FORT=
1TO50:NEXTT:SOUND8,0
780 IFMN=3ANDZ=1THEN800ELSEIFMN=2ANDZ=1
THEN840ELSEIFMN=1ANDZ=1THEN880ELSEIFMN=
3ANDZ=2THEN810ELSEIFMN=2ANDZ=2THEN850EL
SEIFMN=1ANDZ=2THEN890ELSEIFMN=3ANDZ=3TH
EN820ELSEIFMN=2ANDZ=3THEN860ELSEIFMN=1A
NDZ=3THEN900ELSE790
790 IFMN=3ANDZ=4THEN830ELSEIFMN=2ANDZ=4
THEN870ELSEIFMN=1ANDZ=4THEN910ELSE190
800 P1=1:X=4:Y=23:PUTSPRITE1,(X,Y),15,2
:GOTO190
810 P1=1:X=230:Y=62:PUTSPRITE1,(X,Y),15
,4:GOTO190
820 P1=1:X=4:Y=103:PUTSPRITE1,(X,Y),15,
2:GOTO190
830 P1=1:X=230:Y=148:PUTSPRITE1,(X,Y),1
5,4:GOTO190
840 P2=1:X=4:Y=23:PUTSPRITE1,(X,Y),15,2
:GOTO190
850 P2=1:X=230:Y=62:PUTSPRITE1,(X,Y),15
,4:GOTO190
860 P2=1:X=4:Y=103:PUTSPRITE1,(X,Y),15,
2:GOTO190
870 P2=1:X=230:Y=148:PUTSPRITE1,(X,Y),1
5,4:GOTO190
880 P3=1:X=4:Y=23:PUTSPRITE1,(X,Y),15,2
:GOTO190
890 P3=1:X=230:Y=62:PUTSPRITE1,(X,Y),15
:GOTO190
900 P3=1:X=4:Y=103:PUTSPRITE1,(X,Y),15,
2:GOTO190
910 P3=1:X=230:Y=148:PUTSPRITE1,(X,Y),1
5,4:GOTO190
920 SOUND8,0:SOUND1,15:SOUND6,4:SOUND7,
54:SOUND8,16:SOUND12,12:SOUND13,0:FORT=
1TO500:NEXT
940 REM----------GAME OVER----------
960 CLS:COLOR15,1,1:SCREEN1:LOCATE80,50
:PRINT"GAME OVER":LOCATE80,65:PRINT"SCO
RE "SC:SOUND7,60:PLAY"V802T255L4DR64DR6
4DR64L4FR64L15ER64DR64DR64CR64L4D"
970 FORT=1TO200:NEXTT:LOCATE60,100:PRIN
T"PLAY AGAIN ? (Y/N)"
980 R$=INKEY$
990 IFR$="Y"ORR$="y"THEN80
1000 IFR$<>"N"ANDR$<>"n"THEN980
1010 CLS:END
```

```
1030 REM---------DRAW  SCREEN---------
1050 DRAW"C5BM29,0D9R1D10R1D20L31U39":P
AINT(20,10),5:DRAW"C10BM40,40U2R1U2R2D1
R2U2R2D2R2D1R1D2L10":PAINT(42,39),10:D1
$="D2L1D1L1U1L1U2L1U1L3U2L1U1L2D1L1D2L1
U1L2D1L1U2L2":D2$="D2L1D1L1U1L1U2L1U1L3
U2L1U1L2D1L1D2L1U1L2D1L1U2L2D1L1D2L1D1L
1U1L1U2L1U1L3"
1060 DRAW"C10BM40,0D5R4D1R1D2R1D1R1U1R1
U2R1U1R3U2R1U1R2D1R1D2R1U1R2D1L1U2R2D1R
1D2R1D1R1U1R1U2R1U1R3U2R1U3R1U3R20D1R1D
2R2D3R1D1R2D4R4D1R2U4R1U2R3U3R40D2R1D1R
2D1R1D3R3U2R2U1R3U1R1U5":PAINT(41,2),10
:PAINT(100,2),10:PAINT(152,2),10
1070 DRAW"C10BM0,40R230D5L4D1L1XD1$;D2L
4D1L1D2L1D1L1U1L1U2L1U1L3D1L1XD2$;"
1080 DRAW"C10D5L1XD1$;D1L1D2L1D1L1U1L1U
2L1U1L4D1L1XD2$;"
1090 DRAW"C10D5L4XD1$;D1L1D2L1D1L1U1L1U
2L1U1L3L4D5L1XD2$;"
1100 DRAW"C10L4D1L1XD1$;D1L1D9L1D1L1U1L
1U2L1U1L3D2L4D1L1XD2$;"
1110 PAINT(50,41),10
1120 DRAW"C10BM250,80D5L4D1L1XD1$;D1L1D
1L1D1L1U1L1U2L1U1L3D3L4D1L1XD2$;"
1130 DRAW"C10D5L4D1L1XD1$;D1L1D2L1D1L1U
1L1U2L1U1L3L4D1L1XD2$;"
1140 DRAW"C10D5L4D1L1XD1$;D1L1D2L1D1L1U
1L1U2L1U1L3D2L4D1L1XD2$;"
1150 DRAW"C10D5L4D1L1XD1$;D1L1D2L1D1L1U
1L1U2L1U1L3D2L4D1L1D2L1D1L1U1L1U2L1U1L3
U2L1U1L2D1L1U11R230":PAINT(210,82),10
1160 DRAW"C10BM0,120R230D5L4D1L1XD1$;D1
L1D2L1D1L1U1L1U2L1U1L3D2L4D1L1XD2$;"
1170 DRAW"C10D5L4D1L1XD1$;D1L1D2L1D1L1U
1L1U2L1U1L3D2L4D1L1XD2$;"
1180 DRAW"C10D5L4D1L1XD1$;D1L1D2L1D1L1U
1L1U2L1U1L3D2L4D1L1D2L1D1L1U1L1U2L1U1L3
U2L1U1L2D1L1D2L1U1L2D1L1U2D35U35L61":PA
INT(20,121),10:DRAW"C10BM61,155R1U1R1U1
R1U2L1U1L1U1L1"
1190 PAINT(62,153),10
1200 DRAW"C10BM0,165R250D2L75D24L175U26
":PAINT(10,170),10:DRAW"C10BM52,88D32":
DRAW"C10BM51,96R5L10D1L1R12D1R1L14":DRA
W"C10BM24,88D20R1U20":CIRCLE(1,106),2,1
0:PAINT(1,106),10
1210 DRAW"C15BM27,140D1R2D2R2D1R2U1R1D1
R2U1R2U2R2U1D1L2D2L2D4L1D4L1D1L1U1L1U4L
1U2D2R1D4":PAINT(33,147),15
1220 DRAW"C7BM32,155D2L1U1L1U1L1U1L1U1D
1R1D1R1D1R1D1R1D5L3U2L1U2L2D2L1D4R17U4L
1U2L2D2L1D2L3U5R1U1R1U1R1U1R1U1D1L1D1L1
D1L1D1L1U4L1D1L1U1L1D2":PAINT(33,160),7
1230 LOCATE20,170:PRINT"SCORE: ":LOCATE
60,170:PRINTSC
1240 LOCATE100,170:PRINT"PLEASE WAIT"
1250 RETURN
1270 REM--------SET UP SPRITES--------
1290 RESTORE:FORSP=1TO18:SP$=""
1310 FORN=1TO32:READD:SP$=SP$+CHR$(D):N
EXT
1350 SPRITE$(SP)=SP$:NEXT
1370 LINE(99,169)-(172,180),10,BF:RETUR
N
1390 REM---------SPRITES DATA---------
1410 DATA0,0,0,0,0,0,127,31,7,0,0,0,0,0
,0,0,0,0,56,24,12,14,112,112,224,0,0,0,
0,0,0,0:'FLYING RIGHT (1)
1420 DATA0,0,0,0,0,48,24,24,8,4,7,7,3,1
,6,4,24,28,12,4,6,7,4,8,16,32,224,224,1
92,128,64,96:'RUNNING RIGHT(2)
1430 DATA0,0,0,0,0,0,0,0,0,28,24,48,112
,15,15,7,0,0,0,0,0,0,0,0,0,0,0,0,124,
120,224:'DUCKING LEFT(3)
1440 DATA24,56,48,32,96,224,32,16,8,4,7
,7,3,1,2,6,0,0,0,0,12,24,24,16,32,224,2
24,192,128,96,64,0:'RUNNING LEFT(4)
1450 DATA0,0,28,24,48,112,15,15,7,0,0,0
,0,0,0,0,0,0,0,0,0,124,120,224,0,0,0,
0,0,0,0,0:'FLYING LEFT(5)
1460 DATA0,0,0,0,0,0,0,0,0,0,0,0,127,31
,7,0,0,0,0,0,0,0,0,0,56,24,12,14,112,11
2,224,0:'DUCKING RIGHT (6)
1470 DATA0,255,0,0,0,0,0,0,0,0,0,0,0,0,
0,0,8,252,8,0,0,0,0,0,0,0,0,0,0,0,0:'
ARROW RIGHT (7)
1480 DATA35,119,255,255,0,0,0,0,0,0,0,0
,0,0,0,0,128,192,224,240,0,0,0,0,0,0,0,
0,0,0,0,0:'BEETLE LEFT (8)
1490 DATA255,255,255,255,255,255,255,25
5,255,255,255,255,255,255,255,255,0,240,224
,224,192,128,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0:'HANGING ROCK (9)
1500 DATA28,62,127,255,255,255,127,28,0
,0,0,0,0,0,0,0,0,0,0,0,128,128,128,0,0,0,
0,0,0,0,0,0,0:'WATER DROP(10)
1510 DATA64,224,64,64,64,64,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0:
'SWORD(11)
1520 DATA248,248,112,112,32,32,0,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
,0,0:'CRYSTAL (12)
1530 DATA3,3,3,3,3,3,3,3,3,3,3,3,0,0,0,
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0:' BEA
M(13)
1540 DATA0,0,0,63,127,255,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,252,254,255,0,0,0,0,0,0,0,
0,0,0:'BAT(14)
1550 DATA1,7,15,63,127,31,127,31,0,0,0,
0,0,0,0,0,128,208,240,248,240,252,248,2
40,0,0,0,0,0,0,0,0,0:'ROCK  (15)
1560 DATA1,1,33,63,63,3,3,3,3,3,1,1,1,1
,1,1,128,128,132,252,252,192,192,192,19
2,192,128,128,128,128,128,128:'SWORD2
   (16)
1570 DATA 128,192,128,0,0,0,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0:'BALL (17)
1580 DATA60,126,255,255,255,0,0,0,0,0,0
,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
0,0:'ENDING ROCK (18)
1600 REM--------OPENING SCREEN-------
1620 COLOR15,1,1:SCREEN1
1630 LOCATE80,59:PRINT"ROAD RUNNER"
1640 A$="V8T255O4L4CR64L15CR64L15CR64L4
DR64L15DR64L15DR64L4ER64L15ER64L15ER64L
15DR64L3ER64L9ER1"
1650 B$="V8T255O4L4ER64L15ER64L2CR64L15
CR64L4CR64L15DR64L15DR64L4DR64L15CR64L4
DR1"
```

```
1660 PLAYA$+A$:PLAYB$+B$:FORT=1TO2700:N
EXTT
1670 LOCATE20,95:PRINT"PRESS (I) FOR IN
STRUCTIONS";
1680 LOCATE20,104:PRINT"PRESS (P) TO PL
AY";
1690 R$=INKEY$
1700 IFR$="I"ORR$="i"THEN1720
1710 IFR$="P"ORR$="p"THEN80ELSE1690
1720 SCREEN0:COLOR15,1,1:WIDTH40
1730 LOCATE10,1:PRINT"ROAD RUNNER"
1740 LOCATE0,3
1750 PRINT"You are the road runner and
your task isto dodge the coyoty's traps
 and make it to the bottom to destroy t
he coyote."
1760 PRINT"There are four floors each w
ith traps onthem.  If you get hit on an
y floors you are taken back to the star
t of the levelwhere you must try again.
"
1770 PRINT"As you get closer to the end
, your scorebuilds up - at the end of t
he game your score is finalized by givi
ng a bonus forthe amount of men still l
eft."
1780 PRINT"When you jump down to the th
ird level  their is a lever wich you m
ust peck in  order to continue with the
 game.  Also  you must peck the lever a
t the end to  make the anvil fall on t
he coyoty's headGOOD LUCK"
1790 PRINT:PRINT"PRESS SPACE BAR TO CON
TINUE"
1800 R$=INKEY$:IFR$<>" "THEN1800ELSE80
1820 REM------PART OF MAIN LOOP------
1840 IFX>225ANDZ=1THEN1850ELSEIFX<15AND
Z=2THEN1870ELSEIFX>225ANDZ=3THEN1890ELS
EIFX<65ANDZ=4THEN1910ELSERETURN
1850 SOUND8,0:Y=62:X=230:Z=2:NAM=2:LINE
(55,168)-(98,180),10,BF:SC=SC+15:LOCATE
60,170:PRINTSC:PUTSPRITE1,(X,Y),15,P:SO
UND0,0:SOUND6,10:SOUND7,20:SOUND8,16:SO
UND12,30:SOUND13,0
1860 FORT=1TO100:NEXTT:RETURN
1870 SOUND8,0:Y=103:X=4:Z=3:NAM=3:LINE(
55,168)-(98,180),10,BF:SC=SC+20:LOCATE6
0,170:PRINTSC:PUTSPRITE1,(X,Y),15,P:SOU
ND0,0:SOUND6,10:SOUND7,20:SOUND8,16:SOU
ND12,30:SOUND13,0
1880 FORT=1TO100:NEXTT:RETURN
1890 SOUND8,0:Y=148:X=230:Z=4:NAM=4:LIN
E(55,168)-(98,180),10,BF:SC=SC+25:LOCAT
E60,170:PRINTSC:PUTSPRITE1,(X,Y),15,P:S
OUND0,0:SOUND6,10:SOUND7,20:SOUND8,16:S
OUND12,30:SOUND13,0
1900 FORT=1TO100:NEXTT:RETURN
1910 GOTO 2000
1930 REM-----TRAP DOOR SEQUENCE-----
1950 IFY=103ANDFJ=1THEN1960ELSEX=1:GOTO
210
1960 A=92:X=1:LINE(55,168)-(98,180),10,
BF:SC=SC+52:LOCATE60,170:PRINTSC:PLAY"V
8T25506C":FJ=2:GOTO210
1980 REM----LEVER IN LAST PATTERN----
2000 PLAY"V8T25506C":LINE(55,168)-(98,1
80),10,BF:SC=SC+555:LOCATE60,170:PRINTS
C:RF=125
2010 PUTSPRITE23,(30,RF),10,18
2020 RF=RF+1:FORT=1TO50:NEXT:IFRF>135TH
EN2030ELSE2010
2030 SOUND8,0:SOUND1,15:SOUND6,4:SOUND7
,54:SOUND8,16:SOUND12,12:SOUND13,0:FORT
=1TO300:NEXT:GOTO2070
2050 REM----SCENES AFTER COMPLETE----
2070 GOSUB 2160
2080 FORT=1TO1000:NEXTT:CLS:COLOR15,1,1
:SCREEN1:LOCATE80,40:COLOR9:PRINT"//////
//////////"
2090 LOCATE80,49:COLOR10:PRINT"ROAD RUN
NER"
2100 LOCATE80,52:COLOR9:PRINT"//////////
////"
2110 LOCATE60,80:COLOR15:PRINT"BONUS
100XMEN="MN*100
2120 LOCATE60,89:COLOR15:PRINT"
+"
2130 LOCATE60,98:COLOR15:PRINT"SCORE
      "SC
2140 LOCATE60,107:COLOR3:PRINT"TOTAL
      "SC+MN*100
2150 SOUND7,60:PLAY"L805E16DC1604BAG+4L
1B":FORT=1TO2500:NEXTT:GOTO2190
2160 CLS:COLOR15,1,1:SCREEN2
2170 COLOR12:PRINT" WELLDONE"
2180 FORT=1TO1000:NEXTT:GOTO2080
2190 CLS:COLOR15,1,1:SCREEN1
2200 LOCATE60,100:PRINT"PLAY AGAIN ? (Y
/N)"
2210 R$=INKEY$
2220 IFR$="Y"ORR$="y"THEN80
2230 IFR$<>"N"ANDR$<>"n"THEN2210
```

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```
10 ' ================================
20 '
30 '
40 '
50 '    Language: BASIC
60 '    Author  : J.P.Knight
70 '    Date    : June 1986
80 '    Machine : M.S.X
90 ' ================================
100 DEFINTA-Y:KEYOFF:WIDTH32
110 DIM A(24,24)
120 ON INTERVAL=50 GOSUB 2340
130 GOSUB 1670:' Title Screen
140 GOSUB 340:' Get Level
150 GOSUB 430:' Setup Array
160 GOSUB 670:' Draw Screen
170 X=1:Y=19:   ' Initial Sprite Posn
180 INTERVAL ON
190 GOSUB 900:' Position Sprite
200 GOSUB 1070:' Test Outcome
210 '-----------------------------------
220 '     M A I N L I N E
230 '-----------------------------------
240 GOSUB 870:' Get a Move
250 GOSUB 1070:' Test Outcome
260 GOTO 240
270 '-----------------------------------
280 '================================
290 '    S U B - R O U T I N E S
300 '================================
310 '-----------------------------------
320 ' Get Level Of Difficulty
330 '-----------------------------------
340 SCREEN1:WIDTH32:CLS:LOCATE1,1:PRINT
"  PLEASE SELECT DIFFICULTY"
350 LOCATE2,5:PRINT"(1=Easy --> 5=Impos
sible)"
360 LVL$=INKEY$:IF LVL$<"1" OR LVL$>"5"
 THEN 360 ELSE LVL=VAL(LVL$)
370 IF LVL<0 OR LVL>5 THEN 340
380 LOCATE9,21:PRINT"PLEASE WAIT"
390 RETURN
400 '-----------------------------------
410 ' Setup Array With Mines
420 '-----------------------------------
430 ZZ=RND(-TIME):MYNES=50+LVL*20
440 FOR N=1 TO MYNES
450  X=INT(RND(1)*20)+2:Y=INT(RND(1)*20
)+2
460  IF A(X,Y)=1 THEN 450 ELSE A(X,Y)=1
470 NEXT
480 GOSUB 530:' Remove from entrance
490 RETURN
500 '-----------------------------------
510 ' Remove Mines From Entrance
520 '-----------------------------------
530 IF A(2,21) =1 THEN A(2,21)=0:GOSUB
610
540 IF A(3,20) =1 THEN A(3,20)=0:GOSUB
610
550 IF A(21,2) =1 THEN A(21,2)=0:GOSUB
610
560 IF A(20,3) =1 THEN A(20,3)=0:GOSUB
610
570 RETURN
580 '-----------------------------------
590 '  Re-Position
600 '-----------------------------------
610 X=INT(RND(1)*15+4):Y=INT(RND(1)*15+
4)
620 IF A(X,Y)=1 THEN 610 ELSE A(X,Y)=1
630 RETURN
640 '-----------------------------------
650 ' Draw Screen
660 '-----------------------------------
670 COLOR 5,1,6:SCREEN1
680 GOSUB 1430:' Setup User Def Chars
690 GOSUB 1580:' Setup color for chars.
700 FOR N=0 TO 19
710 LOCATE1,N:PRINT STRING$(20,254)
720 NEXT
730 LOCATE22,0:PRINT"MINEFIELD"
740 LOCATE22,1:PRINT"========="
750 LOCATE25,4:PRINT"789"
760 LOCATE25,5:PRINT"4+6"
770 LOCATE25,6:PRINT"123"
780 LOCATE22,9:PRINT"LEVEL:";LVL
790 LOCATE22,12:PRINT"MINES:";MYNES
800 LOCATE24,15:PRINT"TIME"
810 LOCATE3,21:PRINT"MINES IN IMMEDIATE
 AREA:"
820 LOCATE3,22:PRINT"MINES IN   OUTER
 AREA:"
830 RETURN
840 '-----------------------------------
850 ' Get-a-move and Place it
860 '-----------------------------------
870 K$=INKEY$:IF K$="" THEN 870
880 IF K$<"1" OR K$>"9" THEN 870 ELSE K
=VAL(K$)
890 GOSUB2260:ON K GOSUB 960,970,980,99
0,870,1000,1010,1020,1030
900 PUT SPRITE1,(X*8,Y*8),15,1
910 INTERVAL STOP:LOCATEX,Y:PRINT CHR$(
253)
920 RETURN
930 '-----------------------------------
940 '   MOVES
950 '-----------------------------------
960 IF X>1 AND Y<19 THEN X=X-1:Y=Y+1:RE
TURN ELSE RETURN
970 IF Y<19 THEN Y=Y+1:RETURN ELSE RETU
RN
980 IF X<20AND Y<19 THEN X=X+1:Y=Y+1:RE
TURN ELSE RETURN
990 IF X>1  THEN X=X-1:RETURN ELSE RETU
RN
1000 IF X<20 THEN X=X+1:RETURN ELSE RET
URN
1010 IF X>1 AND Y>0  THEN X=X-1:Y=Y-1:R
ETURN ELSE RETURN
1020 IF Y>0  THEN Y=Y-1:RETURN ELSE RET
URN
1030 IF X<20AND Y>0  THEN X=X+1:Y=Y-1:R
ETURN ELSE RETURN
1040 '-----------------------------------
1050 ' Test Outcome Of Move
1060 '-----------------------------------
1070 IF X=20 AND Y=0 THEN 1930:'Success
ful End
1080 X2=X+1:Y2=Y+2:'Array posn.
1090 IF A(X2,Y2)=1   THEN 2030:'Unsucce
ssful End
1100 S=0:L=0:'Short/Long Range
1110 IF A(X2-1,Y2-1)=1 THEN S=S+1
```

```
1120 IF A(X2  ,Y2-1)=1 THEN S=S+1
1130 IF A(X2+1,Y2-1)=1 THEN S=S+1
1140 IF A(X2-1,Y2  )=1 THEN S=S+1
1150 IF A(X2+1,Y2  )=1 THEN S=S+1
1160 IF A(X2-1,Y2+1)=1 THEN S=S+1
1170 IF A(X2  ,Y2+1)=1 THEN S=S+1
1180 IF A(X2+1,Y2+1)=1 THEN S=S+1
1190 ' And now for the long range
1200 IF A(X2-2,Y2-2)=1 THEN L=L+1
1210 IF A(X2-1,Y2-2)=1 THEN L=L+1
1220 IF A(X2  ,Y2-2)=1 THEN L=L+1
1230 IF A(X2+1,Y2-2)=1 THEN L=L+1
1240 IF A(X2+2,Y2-2)=1 THEN L=L+1
1250 IF A(X2-2,Y2-1)=1 THEN L=L+1
1260 IF A(X2+2,Y2-1)=1 THEN L=L+1
1270 IF A(X2-2,Y2  )=1 THEN L=L+1
1280 IF A(X2+2,Y2  )=1 THEN L=L+1
1290 IF A(X2-2,Y2+1)=1 THEN L=L+1
1300 IF A(X2+2,Y2+1)=1 THEN L=L+1
1310 IF A(X2-2,Y2+2)=1 THEN L=L+1
1320 IF A(X2-1,Y2+2)=1 THEN L=L+1
1330 IF A(X2  ,Y2+2)=1 THEN L=L+1
1340 IF A(X2+1,Y2+2)=1 THEN L=L+1
1350 IF A(X2+2,Y2+2)=1 THEN L=L+1
1360 LOCATE26,21:PRINTS;"  "
1370 LOCATE26,22:PRINTL;"  "
1380 INTERVAL ON
1390 RETURN
1400 '-----------------------------------------
1410 ' Setup User Defined Characters
1420 '-----------------------------------------
1430 RESTORE:FOR I=0 TO 7:READ N
1440  VPOKE(BASE(7)+((254*8)+I)),N:NEXT
1450 DATA 255,129,129,129,129,129,129,2
55
1460 FOR I=0 TO 7:READ N
1470  VPOKE(BASE(7)+((253*8)+I)),N:NEXT
1480 DATA 255,255,255,255,255,255,255,2
55
1490 FOR I=0 TO 7:READ N
1500  VPOKE(BASE(7)+((245*8)+I)),N:NEXT
1510 DATA 0,90,60,126,126,60,90,0
1520 MAN$=CHR$(56)+CHR$(56)+CHR$(16)+CH
R$(254)+CHR$(186)+CHR$(186)+CHR$(40)+CH
R$(108)
1530 SPRITE$(1)=MAN$
1540 RETURN
1550 '-----------------------------------------
1560 ' Now Set Colour For U.D.C's
1570 '-----------------------------------------
1580 N=BASE(6)
1590 VPOKE(N+31),&B11100001:' squares
1600 VPOKE(N+30),&B11010001:' mines
1610 VPOKE(N+ 6),&B10000001:' numbers
1620 VPOKE(N+ 7),&B10000001:' numbers
1630 RETURN
1640 '-----------------------------------------
1650 '  Title Screen & Introduction
1660 '-----------------------------------------
1670 SCREEN1:WIDTH31:COLOR10,1
1680 GOSUB1430:PRINT:PRINT
1690 PRINT"▪▪▪▪▪  ▪▪▪  ▪   ▪  ▪▪▪▪▪":GO
SUB 2410
1700 PRINT"▪ ▪ ▪   ▪   ▪▪  ▪  ▪":GOSUB
2410
1710 PRINT"▪ ▪ ▪   ▪   ▪ ▪ ▪  ▪▪▪":GOSU
B 2410
1720 PRINT"▪ ▪ ▪   ▪   ▪  ▪▪  ▪":GOSUB
2410
1730 PRINT"▪ ▪ ▪  ▪▪▪  ▪   ▪  ▪▪▪▪▪":GO
SUB 2410
1740 PRINT:PRINT:PRINT:PRINT
1750 PRINT"▪▪▪▪▪  ▪▪▪  ▪▪▪▪  ▪     ▪▪
▪":GOSUB 2410
1760 PRINT"▪      ▪   ▪     ▪     ▪
 ▪":GOSUB 2410
1770 PRINT"▪▪▪    ▪   ▪▪▪   ▪     ▪
 ▪":GOSUB 2410
1780 PRINT"▪      ▪   ▪     ▪     ▪
 ▪":GOSUB 2410
1790 PRINT"▪      ▪▪▪  ▪▪▪▪  ▪▪▪▪  ▪▪
▪":GOSUB 2410
1800 GOSUB 2610:FORZX=1TO60:GOSUB 2410:
NEXT
1810 SCREEN1:WIDTH31:COLOR10,1
1820 LOCATE6,2:PRINT"M I N E F I E L D"
1830 LOCATE6,3:PRINT"-------------------"
1840 LOCATE5,6:PRINT"A Game Of Strategy
"
1850 LOCATE0,8:PRINT"You are on the edg
e of a mine- field. Your mission is to
get  to the top right corner of the scr
een without stepping on any mines."
1860 PRINT"You are armed with a mine de
t- ector with 2 sensitivities and must
painstakingly plot each   move with the
 aid of your det- ector."
1870 LOCATE5,22:PRINT"HIT ANY KEY TO ST
ART"
1880 K$=INKEY$:IF K$=""THEN 1880
1890 RETURN
1900 '-----------------------------------------
1910 '  Successful End
1920 '-----------------------------------------
1930 GOSUB 2450: 'congratulations
1940 GOSUB 2190:' Print Mine Positions
1950 LOCATE2,21:PRINT"  *** GAME OVER *
**
"
1960 LOCATE2,22:PRINT SPC(28)
1970 DEFUSR0=&H156:Y=USR(0)
1980 K$=INKEY$:IF K$="" THEN 1980 ELSE
BEEP
1990 INTERVAL OFF:HRS=0:MNS=0:SECS=0
2000 LOCATE2,21:PRINT"
          ":GOSUB380
2010 GOSUB2750' CLEAR MINEFIELD
2020 GOTO140
2030 '-----------------------------------------
2040 '  Unsuccessful End
2050 '-----------------------------------------
2060 BEEP:GOSUB 2140:' Explosion
2070 BEEP
2080 GOSUB2560:' Davy Jones
2090 GOSUB 2190:' Print Mine Positions
2100 GOTO 1950
2110 '-----------------------------------------
2120 '  Explosion
2130 '-----------------------------------------
2140 SOUND7,&B10110111:FORI=15TO1STEP-1
:SOUND8,I:FORI2=1TO100:NEXT:NEXT
2150 SOUND7,&B10111111:RETURN
2160 '-----------------------------------------
```

```
2170 ' Print Mine Positions
2180 '-------------------------------
2190 INTERVAL OFF:FOR Y=0 TO 19:FOR X=1
 TO 20
2200 IF A(X+1,Y+2)=1 THEN LOCATEX,Y:PRI
NT CHR$(245)
2210 NEXT:NEXT
2220 RETURN
2230 '-------------------------------
2240 ' Sound On Sprite Move
2250 '-------------------------------
2260 SOUND7,&B10111110:SOUND8,12
2270 FORI=15TO1STEP-2:SOUND1,I:TIME=0
2280 IF TIME<1 THEN 2280
2290 NEXT:SOUND8,0
2300 RETURN
2310 '-------------------------------
2320 ' Timer
2330 '-------------------------------
2340 SECS=SECS+1:IF SECS=60 THEN SECS=0
:MNS=MNS+1
2350 IF MNS=60 THEN MNS=0:HRS=HRS+1
2360 LOCATE22,17:PRINTUSING"##";HRS;:PR
INT":";:PRINTUSING"##";MNS;:PRINT":";:P
RINTUSING"##";SECS
2370 RETURN
2380 '-------------------------------
2390 ' Change color on UDC
2400 '-------------------------------
2410 NY=INT(RND(-TIME)*14+2):NZ=BASE(6)
:VPOKE(NZ+31),16*NY
2420 FOR NZ=1TO300:NEXT:'wait
2430 RETURN
2440 '================================
2450 '--------------------------------
2460 'Congratulations
2470 '--------------------------------
2480 BEEP
2490 A$="116cdeffcccfefggdddfefal22gr64
gfr64fl16edeccc"
2500 B$="o6116cdeffcccfefggdddfefal22gr
64gfr64fl16edeccc"
2510 C$="116cdeffcccfefggdddfefal22gr64
gfr64fl16edefff"
2520 D$="o7116cdeffcccfefggdddfefal22gr
64gfr64fl16edefff"
2530 PLAYA$,B$
2540 PLAYC$,D$
2550 RETURN
2560 '--------------------------------
2570 ' Davy Jones
2580 '--------------------------------
2590 A$="s13m04000t130o3m10001 4dd18d14d
fl8ee.dd.c#14d."
2600 PLAY A$: RETURN
2610 '--------------------------------
2620 ' Yankee doodle
2630 '--------------------------------
2640 A$="o4t25018co6cdecedo3go4co6cdel4
co3b"
2650 D$="o6t25018co4cdecedo5go6co4cdel4
co5b"
2660 B$="o4t25018co6cdefedco3bgabo414co
6c"
2670 E$="o6t25018co4cdefedco5bgabo614co
4c"
2680 C$="o318abagabo4cco3gagfeeggabagab
o4co3ago4co3bo4dcco6cc"
2690 F$="o218abagabo3cco2gagfeeggabagab
o3co5ago6co5bo6dcco4cc"
2700 PLAY A$,D$:PLAY B$,E$:PLAY C$,F$
2710  RETURN
2720 '-------------------------------
2730 ' CLEAR MINEFIELD
2740 '-------------------------------
2750 FORQ=0TO24:FORQQ=0TO24
2760 A(Q,QQ)=0
2770 NEXTQQ:NEXTQ
2780 RETURN
```

WANTED - WANTED - WANTED - WANTED

318 OR 328 CASSETTE RECORDER
80 COLUMN CARD FOR SPECTRAVIDEO 328
SV 902 DISK DRIVE
DISK DRIVE CONTROLLER ETC,ETC

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NOTE CHANGES FOR THE SV-318

```
10 CLEAR1000,53605!
20 ON STOP GOSUB 2800
30 STOP ON
40 SCREEN 0,0:LOCATE,,0:CLS:COLOR 15,1:
WIDTH 39
50 PRINT:PRINT:PRINT
60 PRINT"          N.GREEN PRESENTS "
70 PRINT:PRINT
80 PRINT"        GRAPHICS UNLIMITED"
90 PRINT:PRINT"          §§§§§§§§§§§§§§
§§§ "
100 PRINT:PRINT:PRINT:PRINT:PRINT:PRINT
110 PRINT"        FOR SPECTRAVIDEO SV32
8"
120 PRINT:PRINT
130 PRINT"        Copyright 1985 N.Gree
n"
140 FOR T=1 TO 2000:NEXT T
150 SCREEN 0,0:LOCATE,,0:CLS:COLOR 15,4
:WIDTH 39
160 CLS:PRINT:PRINT:PRINT
170 PRINT"      THIS PROGRAM IS COPYRIG
HT"
180 PRINT" IT MAY BE COPIED AND DISTRIB
UTED FREE"
190 PRINT"        WITHIN AUSTRALASIA,"
200 PRINT"BUT MAY NOT BE SOLD OR INCLUD
ED IN ANY"
210 PRINT"   PACKAGE FOR SALE, NOR SHAL
L THIS"
220 PRINT"     MESSAGE BE REMOVED OR AL
TERED."
230 PRINT" THE AUTHOR IS NOT RESPONSIBL
E FOR ANY"
240 PRINT" RESULTANT USE OF THIS PROGRA
M OR FOR"
250 PRINT"    ANY LOSS CAUSED BY IT'S U
SE."
260 PRINT:PRINT" IF YOU REQUIRE TO PRIN
T A SCREEN AND"
270 PRINT" HAVE ALREADY RUN SD1 THEN PR
ESS <P>"
280 PRINT"        FOR COLOR MENU"
290 PRINT:PRINT" PRESS<X>TO GO IMMEDIAT
ELY TO PROGRAM":PRINT
300 PRINT"    ANY OTHER KEY FOR INSTRUC
TIONS."
310 A$=INKEY$:IF A$=""GOTO 310
320      IF A$="X"OR A$="x"THEN 1090 EL
SE 330
330      IF A$="P" OR A$="p" THEN GOSUB
 3240 ELSE 340
340 CLS:PRINT"          -* COMMAND SUMMARY
 *-":PRINT
350 PRINT"    <ESC> & KEYS 7/8 CLEAR SC
REEN          AND CHANGE BACKGROUND CO
LOR":PRINT
360 PRINT"No.KEYS 1-5 = PEN SPEED/PIXEL
 JUMP":PRINT
370 PRINT" KEYS P&Q = CHANGE PEN COLOR
(+&-)":PRINT
380 PRINT"KEYS I&O ALTER CIRCLE RATIO I
N/OUT":PRINT
390 PRINT" <ENTER> SETS FIRST POINT FOR
 LINE              BOX OR CIRCLE":PRIN
T
400 PRINT" KEYS L,B&C DRAW LINE BOX OR
CIRCLE       FROM <ENTER> POINT TO PEN
TIP":PRINT
410 PRINT" <ESC> THEN <F> FILLS SHAPE W
ITH PEN       COLOR. !!! CAUTION !!!":P
RINT
420 PRINT"    THIS FUNCTION IS UNPREDIC
TABLE."
430 PRINT:PRINT"Press any key to contin
ue"
440 A$=INKEY$:IF A$=""THEN440
450 CLS
460 PRINT"  JOYSTICK #1 MOVES PEN
                  AT SELECTED SPEED
.":PRINT
470 PRINT"  PRESS TRIGGER TO WRITE IN
                    PEN COLOR":PRINT
480 PRINT"    <R> SETS/RESETS TO 'RAY'
MODE":PRINT
490 PRINT"    'R' APPEARS TOP RIGHT WHE
N SET":PRINT
500 PRINT"    <X> TURNS PEN VISIBLE/INV
ISIBLE.":PRINT
510 PRINT"    <*> CENTRES PEN ON SCREEN
":PRINT
520 PRINT:PRINT:PRINT:PRINT
530 PRINT" Press any key to continue"
540 A$=INKEY$:IF A$=""THEN 540
550 CLS
560 PRINT"          -* MIRROR MODES *-":P
RINT
570 PRINT"  IN ANY OF THESE FOUR MODES
ANY       GRAPHIC CREATED AT THE PEN WI
LL ALSO    APPEAR ON THE SCREEN ACCORDI
NG TO             THE MODE SELECTED."
580 PRINT"  IF YOU ARE IN A 'MODE' PRES
SING ANY   OF THE FOLLOWING KEYS WILL R
ESET YOU   TO NORMAL DRAWING.PRESSING A
GAIN WILL            SELECT THAT MODE.
          (RAY MODE IS NOT CHANGED EXC
EPT BY R)":PRINT
590 PRINT"   </> MIRROR IN VERTICAL PL
ANE""
600 PRINT"   <\> MIRROR IN REVERSE PLA
NE"
610 PRINT"   <-> MIRROR IN HORIZONTAL
PLANE"
620 PRINT"   <+> MIRROR FOUR WAYS":PRI
NT
630 PRINT"   INDICATOR APPEARS TOP RIGH
T FOR MODE"
640 PRINT:PRINT:PRINT"Press any key to
continue"
650 A$=INKEY$:IF A$=""THEN 650
660 CLS
670 PRINT"          -* FUNCTION KEYS *-
680 PRINT"           -------------
690 PRINT"   THE FUNCTION KEYS ARE USED
 TO SAVE   AND LOAD SCREENS TO DISK 1:
OR CASS:-"
700 PRINT:PRINT" <F1>TO<F4>  SAVE SCREE
N TO DISK 1 AS     SCREEN1 TO SCREEN4 R
ESPECTIVELY."
```

```
710 PRINT:PRINT" <F6>TO<F9>  LOAD SCREE
N FROM DISK 1 AS      SCREEN1 TO SCREEN4
 RESPECTIVELY."
720 PRINT:PRINT"  SHOULD YOU TRY TO SAV
E TO A FULL DISK  A FLASHING B/W LOGO A
PPEARS TOP LEFT.      PRESS <ESC> TO RE
SUME"
730 PRINT:PRINT"    <F5> SAVES SCREEN TO
 CASSETTE"
740 PRINT" HAVE CASS RECORDING BEFORE <
F5> USE"
750 PRINT"OR'Press play and record' MES
SAGE WILL"
760 PRINT"       BE PRINTED ON THE SCREE
N."
770 PRINT"    <F10> LOADS SCREEN FROM CA
SSETTE"
780 PRINT:PRINT:PRINT"      Press any ke
y to continue"
790 A$=INKEY$:IF A$=""THEN 790
800 CLS: PRINT:PRINT
810 PRINT"          -* TEXT MODE *-":PRIN
T
820 PRINT" TEXT MAY BE PRINTED ON THE S
CREEN IN"
830 PRINT"  EITHER HORIZONTAL OR VERTIC
AL FORM"
840 PRINT"AT THE CURRENT PEN POSITION A
ND COLOR "
850 PRINT:PRINT"    PRESS <T> TO INITIAT
E TEXT MODE"
860 PRINT"  THEN PRESS <V> FOR VERTICAL
 PRINT"
870 PRINT" OR ANY OTHER KEY FOR HORIZON
TAL PRINT"
880 PRINT"     <H> IS RECOMMENDED AS A M
NEMONIC"
890 PRINT:PRINT"    IF PRINT REACHES THE
 BORDER OF THE"
900 PRINT"SCREEN IT WILL RETURN ONE LIN
E DOWN TO"
910 PRINT"  THE STARTING POSITION OF TH
E PEN AT"
920 PRINT" THE <T> COMMAND. SAME FOR <V
>ERTICAL"
930 PRINT"  <ENTER> ALSO DOES THIS FUNC
TION"
940 PRINT:PRINT"  PRESS<ESC>TO RETURN T
O PREVIOUS"
950 PRINT"                MODE"
960 PRINT:PRINT"      Press any key to c
ontinue"
970 A$=INKEY$:IF A$=""THEN 970
980 CLS:PRINT:PRINT:PRINT
990 PRINT"        -* BOLD TEXT MODE *-"
1000 PRINT: PRINT" TEXT MAY ALSO BE IN
SPECIAL FORM IF "
1010 PRINT" THE <B> KEY IS PRESSED AFTE
R <T> AND"
1020 PRINT"   BEFORE <V> OR <H>. THIS F
ORM IS  "
1030 PRINT"          SLIGHTLY WIDER THAN
  "
1040 PRINT"     NORMAL TEXT BUT IS EFFE
CTIVE.":PRINT
1050 PRINT" REMEMBER <T><H> OR <T><V> N
ORMAL TEXT":PRINT
1060 PRINT" <T><B><H> OR <T><B><V> FOR
BOLD TEXT"
1064 PRINT:PRINT" <BACKSPACE> WORKS-<DE
L> DOES NOT"
1065 PRINT:PRINT"     <ESC> TO RESUME D
RAWING"
1070 PRINT:PRINT:PRINT"PRESS ANY KEY TO
 START "
1080 A$=INKEY$:IF A$=""THEN 1080
1090 REM ***** PROGRAM STARTS *****
1100 DEFINT A-Z:DEFSNG S
1110 ON ERROR GOTO 3540
1120 SCREEN1,2:COLOR 15,1,12
1130 REM ***** SET UP PEN SPRITE *****
1140 FOR S=1 TO 16:READ A$,C$:B$=B$+CHR
$(VAL(A$)):D$=D$+CHR$(VAL(C$))
1150 NEXT S
1160 DATA 128,0,120,0,118,0,111,0,95,12
8,63,192,31,224,15,240,7,248,3,252,1,25
4,0,255,0,126,0,60,0,24,0,0
1170 SPRITE$(0)=B$+D$
1180 REM ***** SET UP INDICATOR SPRITES
 *****
1190 FOR S=1 TO 8:READ AA$:BB$=BB$+CHR$
(VAL(AA$))
1200 NEXT S
1210 DATA 254,134,186,134,174,182,186,2
54
1220 SPRITE$(1)=BB$
1230 FOR S=1 TO 8:READ AB$:BC$=BC$+CHR$
(VAL(AB$))
1240 NEXT S
1250 DATA 238,238,238,0,238,238,238,0
1260 SPRITE$(2)=BC$
1270 FOR S=1 TO 8:READ AC$:BD$=BD$+CHR$
(VAL(AC$))
1280 NEXT S
1290 DATA 254,254,254,0,254,254,254,0
1300 SPRITE$(3)=BD$
1310 FOR S=1 TO 8:READ AD$:BE$=BE$+CHR$
(VAL(AD$))
1320 NEXT S
1330 DATA 238,238,238,238,238,238,238,0
1340 SPRITE$(4)=BE$
1350 FOR S=1 TO 8:READ AE$:BF$=BF$+CHR$
(VAL(AE$))
1360 NEXT S
1370 DATA 254,252,250,246,238,222,190,1
26
1380 SPRITE$(5)=BF$
1390 REM ***** SET UP DISK SPRITE *****
1400 DF$="":F$="":FOR S=1 TO 16:READ DF
$,F$:DE$=DE$+CHR$(VAL(DF$)):G$=G$+CHR$(
VAL(F$))
1410 NEXT S
1420 DATA 255,255,254,127,254,127,254,1
27,254,127,255,255,254,127,252,63,252,6
3,254,127,255,255,255,255,127,255,127,2
55,255,255,255,255
1430 SPRITE$(6)=DE$+G$
1440 REM ***** SET UP PEN TIP SPRITE **
***
1450 FOR S=1 TO 8:READ PS$:PT$=PT$+CHR$
(VAL(PS$))
1460 NEXT S
1470 DATA 128,0,0,0,0,0,0,0
```

```
1480 SPRITE$(7)=PT$
1490 REM ***** SET FUNCTION KEYS *****
1500 ON KEY GOSUB 2700,2710,2720,2730,2
740,2750,2760,2770,2780,2790
1510 REM ***** INITIATE FUNCTION KEYS
*****
1520 FOR NN=1TO 10
1530 KEY(NN)ON :NEXT NN
1540  P=15:C=1:Q=15:K=0:R=0:J=0:S=0:M=0
1550 REM **** STARTS HERE AFTER 7 OR 8
****
1560 CLS
1570 IF C>15 THEN C=1 ELSE C=C
1580 IF C<1 THEN  C=15 ELSE C=C
1590 COLOR P,C,12
1600 Z=1:X=128:Y=96
1610 REM **** MAIN LOOP STARTS ****
1620 PUT SPRITE0,(X,Y),P,0
1630 REM ***** FLASHING TIP *****
1640 IF P=0 THEN 1650 ELSE 1660
1650 PUT SPRITE7,(X,Y-1),0,7 :FL=1:GOTO
 1690
1660 IF FL=0 THEN 1670 ELSE 1680
1670 PUT SPRITE7,(X,Y-1),15,7 :FL=1:GOT
O 1690
1680 PUT SPRITE7,(X,Y-1),1,7:FL=0
1690 D=STICK(0)+STICK(1)
1700 V=STRIG(0)+STRIG(1)
1710 IF R>0 GOTO 1750 ELSE1730
1720 REM *** UPDATES COORDS WHEN NOT IN
 RAY MODE ***
1730 E=X:F=Y:E2=X2:F2=Y2
1740 REM ***** UPDATE X1,Y1 IF POINT NO
T SET *****
1750 IF K=0 THEN X1=X:Y1=Y
1760 IF D=1THENY=Y-Z
1770 IF D=2THENY=Y-Z:X=X+Z
1780 IF D=3THENX=X+Z
1790 IF D=4THENX=X+Z:Y=Y+Z
1800 IF D=5THENY=Y+Z
1810 IF D=6THENY=Y+Z:X=X-Z
1820 IF D=7THENX=X-Z
1830 IF D=8THENX=X-Z:Y=Y-Z
1840 IF Y<0 THEN Y=0
1850 IF Y>191 THEN Y=191
1860 IF X<0 THEN X=0
1870 IF X>255 THEN X=255
1880 PUT SPRITE0,(X,Y),P,0
1890 REM *** SET SECOND COORDINATES FOR
 MIRROR MODE ***
1900 X2=127-(X-128):Y2=95-(Y-96)
1910 E2=127-(E-128):F2=95-(F-96)
1920 REM ***** SET <ENTER> POINT COORDS
 FOR M/M *****
1930 X3=127-(X1-128):Y3=95-(Y1-96)
1940 REM ***** READ TRIGGER *****
1950 IF V=-1 GOTO 2460
1960 REM ***** READ KEYBOARD *****
1970 A$=INKEY$
1980      IF A$="" GOTO 1620
1990 REM **** KEY/VARIABLE*** 1-5/Z * 7
&8/C * Q&P/P * R/R * <ENTER>/X1,Y1,K
 * *** I&O/Q * <ESC>/S * X/W * +,-,/,\,
/M *
2000 REM ***** TURN FUNCTION KEYS ON **
***
2010 FOR NN=1TO 10
2020 KEY(NN)ON :NEXT NN
2030       IF VAL(A$)>0 AND VAL(A$)<6 TH
EN Z=VAL(A$)
2040       IF A$="T" OR A$="t" THEN GOSU
B 2830
2050       IF A$="7"THEN IF S=1 THEN C=C
-1:S=0:GOTO 1550
2060       IF A$="8"THEN IF S=1 THEN C=C
+1:S=0:GOTO 1550
2070       IF A$="P"OR A$="p"THEN P=P+1:
IF P>15 THEN P=1
2080       IF A$="Q"OR A$="q"THEN P=P-1:
IF P<1 THEN P=15
2090       IF A$="R"OR A$="r" GOTO 2530
2100       IF A$ =CHR$(13)THEN K=1:X1=X:
Y1=Y:PSET(X1,Y1),P
2110 \     IF A$="L"OR A$="l"THEN LINE(X
1,Y1)-(X,Y),P:K=0
2120       IF A$="L"OR A$="l"THEN IF M=1
 THEN LINE(X1,Y3)-(X,Y2),P
2130       IF A$="L"OR A$="l"THEN IF M=2
 THEN LINE(X3,Y1)-(X2,Y),P
2140       IF A$="L"OR A$="l"THEN IF M=3
 THEN LINE(X3,Y1)-(X2,Y),P:LINE(X1,Y3)-
(X,Y2),P:LINE (X3,Y3)-(X2,Y2),P
2150       IF A$="L"OR A$="l"THEN IF M=4
 THEN LINE(X3,Y3)-(X2,Y2),P
2160       IF A$="B"OR A$="b"THEN LINE(X
1,Y1)-(X,Y),P,B:K=0
2170       IF A$="B"OR A$="b"THEN IF M=1
 THEN LINE(X1,Y3)-(X,Y2),P,B
2180       IF A$="B"OR A$="b"THEN IF M=2
 THEN LINE(X3,Y1)-(X2,Y),P,B
2190       IF A$="B"OR A$="b"THEN IF M=3
 THEN LINE(X3,Y1)-(X2,Y),P,B:LINE(X1,Y3
)-(X,Y2),P,B:LINE (X3,Y3)-(X2,Y2),P,B
2200        IF A$="B"OR A$="b"THEN IF M=
4 THEN LINE(X3,Y3)-(X2,Y2),P,B
2210       IF A$="C"OR A$="c"GOTO 2590
2220 REM ***** CIRCLE RATIO IN OR OUT
*****
2230       IF A$="I"OR A$="i" THEN Q=Q+1
2240       IF A$="O" OR A$="o" THEN Q=Q-
1:IF Q<1 THEN Q=1
2250 REM ***** SAFETY CATCH FOR FILL **
***
2260       IF A$=CHR$(&H1B) THEN S=1
2270 REM ***** FILL SHAPE WITH PEN COLO
R *****
2280       IF A$="F"OR A$="f"THEN IF S=1
 THEN IF M=0 THEN PAINT(X,Y),P:S=0
2290       IF A$="F"OR A$="f"THEN IF S=1
 THEN IF M=1 THEN PAINT(X,Y2),P:PAINT(X
,Y),P:S=0
2300       IF A$="F"OR A$="f"THEN IF S=1
 THEN IF M=2 THEN PAINT(X2,Y),P:PAINT(X
,Y),P:S=0
2310       IF A$="F"OR A$="f"THEN IF S=1
 THEN IF M=3 THEN PAINT(X,Y2),P:PAINT(X
2,Y),P:PAINT(X2,Y2),P:PAINT(X,Y),P:S=0
2320        IF A$="F"OR A$="f"THEN IF S=
1 THEN IF M=4 THEN PAINT(X2,Y2),P:PAINT
(X,Y),P:S=0
2330       IF A$="X"OR A$="x"GOTO 2670
2340       IF A$="*"THEN X=128:Y=96
```

```
2350       IF A$="-"THEN IF M>0 THEN M=0
 ELSE M=1
2360       IF A$="/"THEN IF M>0 THEN M=0
 ELSE M=2
2370       IF A$="+"THEN IF M>0 THEN M=0
 ELSE M=3
2380       IF A$="\"THEN IF M>0 THEN M=0
 ELSE M=4
2390 REM ***** TURN INDICATORS ON/OFF *
****
2400 IF M=1 THEN PUT SPRITE3,(247,0),P,
3 ELSE PUT SPRITE3,(247,-32),0,3
2410 IF M=2 THEN PUT SPRITE4,(247,0),P,
4 ELSE PUT SPRITE4,(247,-32),0,4
2420 IF M=3 THEN PUT SPRITE2,(247,0),P,
2 ELSE PUT SPRITE2,(247,-32),0,2
2430 IF M=4 THEN PUT SPRITE5,(247,0),P,
5 ELSE\PUT SPRITE5,(247,-32),0,5
2440 GOTO 1620
2450 REM ***** DRAW *****
2460 IF M>0 GOTO 2470 ELSE PSET (X,Y),P
:LINE(E,F)-(X,Y),P :GOTO 1970
2470 IF M>1 GOTO 2480 ELSE PSET (X,Y),P
:LINE(E,F)-(X,Y),P:PSET(X,Y2),P:LINE(E,
F2)-(X,Y2),P:GOTO 1970
2480 IF M>2 GOTO 2490 ELSE PSET (X,Y),P
:LINE(E,F)-(X,Y),P:PSET(X2,Y),P:LINE(E2
,F)-(X2,Y),P:GOTO 1970
2490 IF M>3 GOTO 2500 ELSE PSET (X,Y),P
:LINE(E,F)-(X,Y),P:PSET(X2,Y),P:LINE(E2
,F)-(X2,Y),P:PSET(X2,Y2),P:LINE(E2,F2)-
(X2,Y2),P:PSET(X,Y2),P:LINE(E,F2)-(X,Y2
),P:GOTO 1970
2500  IF M>4 GOTO 2510 ELSE PSET (X,Y),
P:LINE(E,F)-(X,Y),P:PSET(X2,Y2),P:LINE(
E2,F2)-(X2,Y2),P:GOTO 1970
2510 M=0:GOTO 1970
2520 REM ***** TOGGLE FOR RAY MODE ****
*
2530 IF R=0 THEN R=1 ELSE R=0
2540 IF R=1 THEN PSET(X,Y),P
2550 REM ***** TURN INDICATOR ON/OFF **
***
2560 IF R=1 THEN PUT SPRITE1,(238,0),P,
1 ELSE PUT SPRITE1,(238,-32),0,1
2570 GOTO 1620
2580 REM ***** CALCULATE RADIUS & DRAW
CIRCLE ******
2590 D= INT(SQR((X1-X)^2+(Y1-Y)^2))
2600 CIRCLE (X1,Y1),D,P,,,Q/12:PSET(X1,
Y1),C:K=0
2610 IF M=1THEN CIRCLE (X1,Y3),D,P,,,Q/
12:PSET(X1,Y3),C:K=0
2620 IF M=2THEN CIRCLE (X3,Y1),D,P,,,Q/
12:PSET(X3,Y1),C:K=0
2630 IF M=3THEN CIRCLE (X3,Y1),D,P,,,Q/
12:CIRCLE(X1,Y3),D,P,,,Q/12:CIRCLE(X3,Y
3),D,P,,,Q/12
2640  IF M=4THEN CIRCLE (X3,Y3),D,P,,,Q
/12:PSET(X3,Y3),C:K=0
2650 GOTO 1620
2660 REM ***** TOGGLE PEN VISIBLE/INVIS
IBLE *****
2670 IF P=0 THEN P=W :GOTO 1620
2680 IF P>0 THEN W=P:P=0:GOTO 1620
2690 END
2700 SAVE "1:SCREEN1",S:RETURN 1620
2710 SAVE "1:SCREEN2",S:RETURN 1620
2720 SAVE "1:SCREEN3",S:RETURN 1620
2730 SAVE "1:SCREEN4",S:RETURN 1620
2740 CSAVE "SCREEN",S:RETURN 1620
2750 LOAD"1:SCREEN1":RETURN 1620
2760 LOAD"1:SCREEN2":RETURN 1620
2770 LOAD"1:SCREEN3":RETURN 1620
2780 LOAD"1:SCREEN4":RETURN 1620
2790 CLOAD"SCREEN":RETURN 1620
2800 SCREEN 0,1:COLOR 15,4,4:LOCATE,,1
2810 END
2820 REM ***** TEXT MODES *****
2830 A$=INKEY$:IF A$=""THEN 2830 ELSE I
F A$="V"OR A$="v"THEN GOSUB 2930 ELSE I
F A$="B"OR A$="b"THEN 3030 ELSE 2840
2840 T=X
2850 A$=INKEY$
2860 IF A$=""THEN 2850
2870 IF A$=CHR$(&H1B)THEN T=0:RETURN 16
20
2880 IF A$=CHR$(8)THEN IF X=>(T+7) THEN
 X=X-7 :GOTO 2900:ELSE 3000
2890 LOCATE X,Y:COLOR P:PRINT A$:X=X+7:
IF A$=CHR$(13) OR X>248 THEN X=T:Y=Y+8
2900 PUT SPRITE0,(X,Y),P,0
2910 GOTO 2850
2920 END
2930 A$=INKEY$
2940 T=Y
2950 A$=INKEY$
2960 IF A$=""THEN 2950
2970 IF A$=CHR$(&H1B)THEN T=0: RETURN 1
620
2980 IF A$=CHR$(8)THEN IF Y=>(T+8) THEN
 Y=Y-8 :GOTO 3000:ELSE 3000
2990 LOCATE X,Y:COLOR P:PRINT A$:Y=Y+8:
IF A$=CHR$(13) OR Y>184 THEN Y=T:X=X+7
3000 PUT SPRITE0,(X,Y),P,0
3010 GOTO 2950
3020 END
3030 T=X
3040 A$=INKEY$
3050 IF A$=""THEN 3040 ELSE IF A$="V"OR
 A$="v"THEN 3130 ELSE 3060
3060 A$="":A$=INKEY$:IF A$=""THEN 3060
3070 IF A$=CHR$(&H1B)THEN T=0: RETURN 1
620
3080 IF A$=CHR$(8)THEN IF X=>(T+8) THEN
 X=X-8 :GOTO 3100:ELSE 3100
3090 LOCATE X-1,Y:COLOR P:PRINT A$: LOC
ATE X,Y:COLOR P:PRINT A$:X=X+8:IF X>248
 OR A$=CHR$(13) THEN X=T:Y=Y+8
3100 PUT SPRITE0,(X,Y),P,0
3110 GOTO 3060
3120 END
3130 A$=INKEY$
3140 T=Y
3150 A$=INKEY$
3160 IF A$=""THEN 3150
3170 IF A$=CHR$(&H1B)THEN T=0: RETURN 1
620
3180 IF A$=CHR$(8)THEN IF Y=>(T+9) THEN
 Y=Y-9 :GOTO 3200:ELSE 3200
```

```
3190 LOCATE X-1,Y:COLOR P:PRINT A$: LOC
ATE X,Y:COLOR P:PRINT A$:Y=Y+9:IF Y>184
 OR A$=CHR$(13) THEN Y=T:X=X+8
3200 PUT SPRITE0,(X,Y),P,0
3210 GOTO 3150
3220 END
3230 REM ***** PRINTER COLOR TO PRINT *
****
3240 WIDTH39:COLOR 15,1
3250 'the list of colours to print star
ts at 53607. Remember to fill unused
  spaces with 255 so as not to print un
wanted colours.
3260 CLS:WIDTH 39:COLOR 15,1
3270 DIM B(16):FOR N=1 TO 15: B(N)=255:
NEXT N
3280 PRINT" WHEN FINISHED <SELECT> BUTT
ON WILL"
3290 PRINT"    PRINT LARGE AND <PRINT>
SMALL"
3300 PRINT
3310 PRINT"          -* COLOR NUMBERS *-"
3320 PRINT"1-BLACK      6-DARK RED    11-
LT YELLOW"
3330 PRINT"2-MED GREEN 7-CYAN         12-
DARK GREEN"
3340 PRINT"3-LT GREEN  8-MED RED      13-
MAGENTA"
3350 PRINT"4-DARK BLUE 9-LT RED       14-
GREY"
3360 PRINT"5-LT BLUE   10-DK YELLOW 15-
WHITE"
3370 PRINT
3380 PRINT" ENTER THE NUMBERS OF ANY CO
LORS "
3390 PRINT"      THAT YOU WANT PRINTED."
3400 PRINT"    ENTER <0> WHEN FINISHED"
3410 FOR Z=1 TO 15
3420 INPUT I
3430 IF I=0THEN GOTO 3470
3440 IF I>15 THEN PRINT"TOO HIGH":GOTO
3420
3450 B(I)=I
3460 NEXT Z
3470 W=0
3480 FOR XT=53607! TO 53621!
3490 W=W+1
3500 POKE XT,B(W)
3510 NEXT
3520 RETURN 160
3530 REM ***** ERROR RECOVER ROUTINE **
***
3540 IF ERR <> 68 THEN RESUME1620 ELSE
3550
3550 A$=INKEY$:IF A$ = CHR$(&H1B) THEN
3600 ELSE 3560
3560 PUT SPRITE6,(8,8),15,6
3570 FOR T=1 TO 100
3580 PUT SPRITE6,(8,8),1,6
3590 FOR T=1 TO 100:GOTO 3550
3600 PUT SPRITE6,(8,8),0,6:RESUME 1620
3610 REM ***** FOR SV-318 *****
3620 REM DELETE ",53605" LINE 10
3630 REM DELETE LINES 260-280
3640 REM DELETE LINES 3230-3520
```

SYNTAX ERRORS

There is a small error in MSX program named,

"REAL MARTIANS DON'T EAT APPLES" 2 lines are wrong they are-

50 The screen command should be SCREEN 2

60 The screen command should be SCREEN 2,2

WE APPOLOGIZE FOR ANY INCONVEINIENCE.

```
10 ' /-------------------------------\
11 ' |HOME BANKING MANAGER
12 ' |by Roger Stockburger
13 ' |
14 ' |
15 ' |      V1.0 WRITTEN 25FEB86
16 ' |
17 ' |
18 ' |
19 ' |   GRAPHICS PRINTOUTS FROM THIS
20 ' |   PROGRAM ARE COMPATIBLE  ONLY
21 ' |   WITH MSX X-Y PLOTTERS,  LIKE
22 ' |   THE SONY PRN-C41 OR SIMILAR.
23 ' \-------------------------------/
24 '
25 ' /-------------------------------<<<
26 ' \---> PROGRAM STARTS HERE -\
27 '                           |
28 '
40 CLEAR 2000,55000!
50 PI$=""
60 MAXFILES=2
70 ON STOP GOSUB 3760:STOPON
80 STOP ON:DIM MF(32):DIM T0(32),T1(32)
,D1(32),W1(32),GR(32),TQ$(100)
100 ON ERROR GOTO 9100:ON STOP GOSUB 37
60:STOPON
105 SCREEN0:WIDTH40:COLOR15,1:KEYOFF:CL
S:BEEP
110 CLOSE:PRINT"/----------------------
----------------\|====== MSX HOME BANKI
NG MANAGER ======|";
120 PRINT"|[C]1986 ROGER STOCKBURGER MS
X SOFTWARE|\--------------------------
-----------/"
130 PRINT"MAIN MENU:":PRINT"==========="
:PRINT:PRINT"[1] - BANKING TRANSACTIONS
":PRINT"      and GRAPHICS MODE":PRINT"
     * Accessed from this mode":PRINT
140 PRINT"[2] - CLEAR MEMORY AND RE-STA
RT PROGRAM":PRINT:PRINT"[3] - SELECT NA
ME OF BANK":PRINT:PRINT"[4] - EXIT TO B
ASIC"
150 LOCATE 0,19:PRINT"================
========================PRESS [1][2][3][
4] KEYS TO SELECT OPTION==============
========================="
160 IF INKEY$<>"" THEN 160
170 R$=INKEY$:IF R$="" THEN 170
180 IF R$="1" THEN 1000 ELSE IF R$="2"
THEN 7990 ELSE IF R$="3" THEN 9000 ELSE
 IF R$="4" THEN 400 ELSE 170
400 COLOR 15,4,4:WIDTH 37:BEEP:CLS:PRIN
T"MSX BASIC V1.0":DEFUSR1=&H9F:X=USR1(1
)
410 NEW
1000 ' * BANKING MANAGEMENT ROUTINES *
1005 KEYOFF:PX=0
1010 IF PI$="" THEN 7000 ELSE CLS:PRINT
"/--------- BANKING TRANSACTIONS -------
-\|";TAB(39);"|"
1020 PRINT"Options available:":PRINT"--
---------------":PRINT:PRINT:PRINT"[1]
- ENTER NEW TRANSACTION":PRINT:PRINT"[2
] - EXAMINE TRANSACTIONS"
1030 PRINT:PRINT:PRINT"PRESS KEY OF CHO
ICE REQUIRED"
1040 PRINT:PRINT"|";TAB(39);"|":PRINT"
\--------------------------------------
/"
1050 IF INKEY$<>"" THEN 1050
1060 A$=INKEY$:IF A$="1" THEN 1070 ELSE
 IF A$="2" THEN 2000 ELSE 1060
1070 CLS:PRINT"/--------- ENTER TRANSAC
TION: ---------\|";TAB(39);"|"
1080 LOCATE 0,3:INPUT"* DATE OF TRANSAC
TION (DD/MM/YYYY)     ";TD$:IF LEN(TD$)<
>10 THEN LOCATE 0,4:PRINT SPC(30):GOTO
1080
1090 TR$="":LOCATE 0,6:INPUT"* TRANSACT
ION DETAILS: 39 LETTERS MAX.";TR$:IF LE
N(TR$)>39 THEN LOCATE 0,7:PRINT SPC(80)
:GOTO 1090
1100 ZZ=0:TR$=" "+TR$:Z=LEN(TR$):FOR ZZ
=Z TO 40:TR$=TR$+" ":NEXTZZ:TR$=LEFT$(T
R$,40)
1110 LOCATE 0,10:PRINT"* TRANSACTION AM
OUNT:":PRINT"  [D] - DEPOSIT     [W] - W
ITHDRAWAL     ---------- PRESS DESIRED
KEY: ----------"
1120 A$=INKEY$:IF A$="W" OR A$="w" THEN
 NG=1:GOTO 1130 ELSE IF A$="D" OR A$="d
" THEN NG=0:GOTO 1130 ELSE 1120
1130 LOCATE 0,11:PRINT SPC(80):LOCATE 0
,11:INPUT "  -> AMOUNT: ";TA:IF TA<0 TH
EN 1130 ELSE IF NG=1 THEN TA=-TA
1135 GOTO 1210
1140 TA$=STR$(TA):Z=LEN(TA$):FOR ZZ=Z T
O 15:TA$=TA$+" ":NEXT:TA$=LEFT$(TA$,16)
1150 GOSUB 1900:IF RT=0 THEN QQ=0:TB=0:
GOTO 1180
1160 ' FIND LAST ACCT. BAL
1170 TB=0:TS$="":OPEN FI$ AS #1:FIELD #
1,80 AS TS$:GET #1,RT:QQ$=MID$(TS$,57,1
6):TB=VAL(QQ$):CLOSE
1180 TB=TB+TA:TB$=STR$(TB):Z=LEN(TB$):F
OR ZZ=Z TO 15:TB$=TB$+" ":NEXT:TB$=LEFT
$(TB$,16)
1190 TX$=MID$(TD$,1,2)+MID$(TD$,4,2)+MI
D$(TD$,7,4):TD$=TX$
1200 TT$=TR$+TA$+TB$+TD$:CX$=MID$(TD$,5
,4):CO=VAL(CX$)
1205 GOTO 1230
1210 PRINT:PRINT:PRINT"================
========================IS THE ABOVE IN
FORMATION CORRECT [Y]/[N]=============
========================="
1220 A$=INKEY$:IF A$="" THEN 1220 ELSE
IF A$="Y" OR A$="y" THEN 1140 ELSE IF A
$="N" OR A$="n" THEN 1000 ELSE 1220
1230 ' WRITE FILE TO DISK
1240 OPEN FI$ AS #1:OPEN FC$ FOR APPEND
 AS #2
1250 FIELD #1,80 AS TS$
1260 RT=RT+1:LSET TS$=TT$:PUT #1,RT:PRI
NT#2,CO
1270 CLS:PRINT"/-----------------------
---------------\| THE FILE YOU ENTERED
IS NOW ON DISK. |!";TAB(39);"|| OPTIONS
:";TAB(39);"|| [1] - ENTER ANOTHER TRAN
SACTION";SPC(6);"|| [2] - GOTO MAIN MEN
U";
```

```
1280 PRINT SPC(17);"|| [3] - GOTO TRANS
ACTIONS SUB-MENU      |\----------------
----------------------/":CLOSE:PRINT:PR
INT"====  PRESS KEY OF CHOICE DESIRED:
 ===="
1290 A$=INKEY$:IF A$="" THEN 1290 ELSE
IF A$="1" THEN 1070 ELSE IF A$="2" THEN
 100 ELSE IF A$="3" THEN 1000 ELSE 1290
1890 END
1900 CLOSE
1910 RT=0:OPEN FC$ FOR INPUT AS #1
1915 IF EOF(1)=-1 GOTO 1930
1920 INPUT#1,C:RT=RT+1:GOTO 1915
1930 CLOSE:RETURN
2000 ' * EXAMINE TRANSACTIONS *
2005 KEYOFF:PX=0
2010 CLS:PRINT"/-------- EXAMINE TRANSA
CTIONS --------\|";TAB(39);"|":PRINT:PR
INT"OPTIONS AVAILABLE:":PRINT"=========
=========":PRINT:PRINT"[1] - SEARCH TRA
NSACTIONS";SPC(15);"[2] - DISPLAY ALL T
RANSACTION IN SEQ."
2015 ON STOP GOSUB 3760:STOPON
2020 PRINT"[3] - GRAPHICS MODE: SCREEN"
:PRINT"[4] - GRAPHICS MODE: MSX X-Y PLO
TTER"
2030 PRINT"[5] - GOTO MAIN MENU":PRINT:
PRINT"|";TAB(39);"|\------------------
-------------------/":PRINT:PRINT"=====
 PRESS KEY OF CHOICE DESIRED: ====="
2040 A$=INKEY$:IF A$="" THEN 2040 ELSE
IF ASC(A$)<49 OR ASC(A$)>53 THEN 2040
2050 A=VAL(A$):ON A GOTO 3055,3600,5100
,6000,100
3000 ' * GET MONTH SUBROUTINE *
3010 INPUT M$:IF LEN(M$)<>3 THEN GOTO 3
000
3020 IF M$="JAN" THEN MN$="01" :GOTO 30
50 ELSE IF M$="FEB" THEN MN$="02" :GOTO
 3050 ELSE IF M$="MAR" THEN MN$="03" :G
OTO 3050 ELSE IF M$="APR" THEN MN$="04"
 :GOTO 3050 ELSE IF M$="MAY" THEN MN$="
05" :GOTO 3050 ELSE IF M$="JUN" THEN MN
$="06" :GOTO 3050
3030 IF M$="JUL" THEN MN$="07" :GOTO 30
50 ELSE IF M$="AUG" THEN MN$="08" :GOTO
 3050 ELSE IF M$="SEP" THEN MN$="09" :G
OTO 3050 ELSE IF M$="OCT" THEN MN$="10"
 :GOTO 3050 ELSE IF M$="NOV" THEN MN$="
11" :GOTO 3050 ELSE IF M$="DEC" THEN MN
$="12" :GOTO 3050
3040 GOTO 3000
3045 VM$=M$
3050 VM$=M$:RETURN
3055 KEYOFF:GOSUB 3800:CLS:PRINT"------
------ SEARCH TRANSACTION ----------":PRI
NT:PRINT"ENTER MONTH TO BE FOUND:
          (JAN,FEB,MAR ABBREVIATION...)
":GOSUB 3000:CLOSE:FOR A=0 TO 31:MF(A)=
0:NEXT:GOSUB 1900
3060 OPEN FI$ AS #1:FIELD#1,80 AS TS$:M
M=1:FOR Z=N TO RT:GET#1,Z:IF MID$(TS$,7
5,2)<>MN$ AND MD=1 OR VAL(RIGHT$(TS$,4)
)<>YY THEN NEXT:GOTO 3080 ELSE IF MID$(
TS$,75,2)=MN$ THEN MF(MM)=Z:MM=MM+1:MD=
1:NEXT Z:GOTO 3080
3070 NEXT Z
3080 ON STOP GOSUB 3760:STOPON
3090 KEY1,"RETURN=":KEY2,"Next":KEY3,"
     ":KEY4,"  [P]=":KEY5,"Print":KEY
ON:CLS:PRINT"TRANSACTIONS: ON FILE ";FI
$:PRINT:PRINT"PRESS ANY KEY TO VIEW TRA
NSACTIONS...":PRINT:PRINT
3100 CLOSE:OPEN FI$ AS #1:FIELD#1,80 AS
  TS$:FOR A=1 TO Z:IF MF(A)=0 THEN 3120
 ELSE GET#1,MF(A)
3110 GOSUB 3980:PRINT MID$(TS$,73,2)+"/
"+MID$(TS$,75,2)+"/"+MID$(TS$,77,4);"
     ";"TR. $";MID$(TS$,41,16):PRINT SP
C(17);"BAL. $";MID$(TS$,57,16):PRINT MI
D$(TS$,1,40):GOSUB 3980:GOSUB 3990:IF R
T=1 THEN 3120
3115 NEXT
3120 DEFUSR1=&H9F:X=USR1(1):GOTO 1000
3600 CLS:PRINT"/- VIEW ALL TRANSACTIONS
 IN SEQUENCE: -\|";TAB(39);"|":PRINT"==
====================================R
ETURN:Next  CTRL STOP:Quit [P]-PRINTOUT
";
3605 PRINT"============================
============"
3610 PRINT:PRINT"[1] - START LISTING TR
ANSACTIONS":PRINT"[2] - HARD COPY OF AL
L TRANSACTIONS":ON STOP GOSUB 3760:STOP
 ON
3615 KEYON:KEY1,"[RET] -":KEY2,"Next":K
EY 3,"<CTRL>":KEY4,"<STOP>-":KEY 5,"EXI
T"
3620 A$=INKEY$:IF A$="" THEN 3620 ELSE
IF A$="1" THEN 3630 ELSE IF A$="2" THEN
 6700 ELSE BEEP:GOTO 3620
3630 CLS:GOSUB 1900:CLOSE:OPEN FI$ AS #
1:FIELD#1,80 AS TS$:FOR Z=1 TO RT:GET #
1,Z
3640 GOSUB 3980:PRINT MID$(TS$,73,2)+"/
"+MID$(TS$,75,2)+"/"+MID$(TS$,77,4);"
     ";"TR. $";MID$(TS$,41,16):PRINT SP
C(17);"BAL. $";MID$(TS$,57,16):PRINT MI
D$(TS$,1,40):GOSUB 3980
3650 A$=INKEY$:IF A$="" THEN 3650 ELSE
IF A$=CHR$(13) THEN NEXT Z ELSE IF A$="
Q" OR A$="q" THEN 2000 ELSE 3650
3660 PRINT:PRINT"========== ALL FILES DI
SPLAYED: ==========":PRINT"==========----
 PRESS ANY KEY ----------":DEFUSR1=&H
9F:X=USR1(1):GOTO 2000
3760 CLS:LOCATE 0,10:PRINT"------------
 A B O R T E D ------------":PRINT:PRIN
T"PRESS ANY KEY TO GOTO MENU":DEFUSR1=&
H9F:X=USR1(1):STOPOFF:RETURN 100
3800 CLS:PRINT"========== SEARCH TRANSA
CTION ==========":PRINT:PRINT" ENTER YE
AR OF TRANSACTIONS REQUIRED:":PRINT:INP
UT "Enter as YYYY (i.e. 1987):";YY:GOTO
 3820
3810 GOSUB 1900:OPEN FC$ FOR INPUT AS #
1:FOR Z=1 TO RT:INPUT #1,CD:IF CD=YY TH
EN 3815 ELSE NEXT:CLOSE:PRINT:PRINT"***
 NO TRANSACTIONS FOR ";YY;" ***":PRINT:
PRINT"->PRESS ANY KEY TO CONTINUE->":DE
FUSR1=&H9F:X=USR1(1):GOTO 2000
3815 RETURN
3820 GOSUB 3810:N=Z:CLOSE:RETURN
3900 '
```

```
3910 LPRINTMID$(TS$,73,2);"/";MID$(TS$,
75,2);"/";RIGHT$(TS$,4);SPC(3);"Transac
t $";MID$(TS$,41,16):LPRINT SPC(13);"Ba
lance  $";MID$(TS$,57,16):LPRINT LEFT$(
TS$,40):LPRINT:RETURN
3980 PRINT"=============================
============";:RETURN
3990 ' * REJ/PRINT SUBMENU *
3995 A$=INKEY$:IF A$="" THEN 3995 ELSE
IF A$=CHR$(13) THEN RETURN ELSE IF A$="
P" OR A$="p" THEN 3900 ELSE 3995
4000 ' GRAPHICS: SELECT SCREEN/PRINTER
5000 FL=0:FOR ZZ=1 TO TQ:IF VAL(MID$(TQ
$(ZZ),73,2))=Z AND VAL(RIGHT$(TQ$(ZZ),4
))=YY AND MID$(TQ$(ZZ),75,2)=MN$ THEN R
ETURN ELSE NEXT ZZ:FL=1:RETURN
5020 LOCATE 0,4:PRINT"[1] - ACCOUNT BAL
ANCES":PRINT"[2] - DEPOSITS":PRINT"[3]
- WITHDRAWALS":PRINT:PRINT"The above gr
aphs plotted by month only":PRINT:PRINT
"PLEASE SELECT GRAPH TO BE SELECTED"
5030 A$=INKEY$:IF A$="" THEN 5030 ELSE
IF ASC(A$)<49 OR ASC(A$)>51 THEN 5030
5035 ZX$=A$
5038 IF PX=1 THEN 5060
5040 LOCATE 0,4:PRINTSPC(240);SPC(80):L
OCATE 0,4:PRINT"PLEASE SELECT TYPE OF G
RAPH:":PRINT:PRINT"[1] - BARGRAPH":PRIN
T"[2] - 'LINE' CHART":PRINT:PRINT"Pleas
e press Key of desired choice:"
5050 AA$=INKEY$:IF AA$="" THEN 5050 ELS
E IF AA$<>"1" AND AA$<>"2" THEN 5050
5060 LOCATE 0,4:PRINT SPC(240);SPC(80):
LOCATE 0,4:INPUT"Please enter YEAR of g
raph: ";YY:LOCATE 0,4:PRINTSPC(240):GOS
UB 3820:LOCATE 0,4:PRINT"Please enter M
ONTH of graph:":PRINT"(Abbreviated like
 JAN, FEB, etc.)":GOSUB 3000
5070 RETURN
5080 X=VAL(MN$):IF X=1 THEN DM=31 ELSE
IF X=2 THEN DM=28 ELSE IF X=3 THEN DM=3
1 ELSE IF X=4 THEN DM=30 ELSE IF X=5 TH
EN DM=31 ELSE IF X=6 THEN DM=30 ELSE IF
 X=7 OR X=8 THEN DM=31 ELSE IF X=9 THEN
 DM=30 ELSE IF X=10 THEN DM=31 ELSE IF
X=11 THEN DM=30
5090 IF X=12 THEN DM=31
5095 RETURN
5100 CLS:PRINT"***** GRAPHICS MODE  (SC
REEN) MENU *****"
5110 GOSUB 5020
5130 CLOSE:GOSUB 1900:OPEN FI$ AS #1:FI
ELD #1,80 AS TS$:FOR Z=1 TO RT:GET #1,Z
:IF VAL(RIGHT$(TS$,4))=YY THEN N=Z:GOTO
 5140 ELSE NEXT:IF Z=1 THEN PRINT"** FI
LE NOT FOUND **":PRINT:PRINT:PRINT"Pres
s any key to continue":DEFUSR1=&H9F:X=U
SR1(1):GOTO 2000
5140 FOR Z=N TO RT:GET#1,Z:IF MID$(TS$,
75,2)=MN$ THEN 5150 ELSE NEXT Z:PRINT"*
* NO TRANSACTION FOUND AS REQUESTED **"
:PRINT:PRINT:PRINT"Press any Key to con
tinue":DEFUSR1=&H9F:X=USR1(1):GOTO 2000
:ELSE 5150
5150 N=Z:Z1=0:GOSUB 5080:FOR Z=N TO RT:
GET#1,Z:IF MID$(TS$,75,2)<>MN$ THEN TQ=
Z1:GOTO 5160 ELSE Z1=Z1+1:TQ$(Z1)=TS$:N
EXT:TQ=Z1
5160 FOR Z=1 TO 31:T0(Z)=0:NEXT
5170 FOR Z=1 TO DM:GOSUB 5000:IF FL=1
THEN T0(Z)=T0(Z-1):NEXT Z ELSE T0(Z)=VA
L(MID$(TQ$(ZZ),57,16)):NEXT Z
5180 CLS:PRINT"Please select scale to b
e used:":PRINT:PRINT"[1] - $0  - $1000"
:PRINT"[2] - $0  - $10000":PRINT"[3] -
$0  - $100000":PRINT"[4] - $0  - $10000
00":PRINT:PRINT"PRESS KEY OF DESIRED CH
OICE"
5190 A$=INKEY$:IF A$="" THEN 5190 ELSE
IF ASC(A$)<49 AND ASC(A$)>52 THEN 5190
ELSE SC=VAL(A$)
5195 IF PX=1 THEN 6020
5200 CLOSE:IF SC=1 THEN DV=7 ELSE IF SC
=2 THEN DV=70 ELSE IF SC=3 THEN DV=700
ELSE IF SC=4 THEN DV=7000
5205 IF ZX$="3" OR ZX$="2" THEN 5500
5206 IF PX=1 THEN 6040
5210 STOPOFF:GOTO 5225
5220 SCREEN 2:COLOR15,1,1:CLS:OPEN"GRP:
" FOR OUTPUT AS #1:PRESET(11,162):COLOR
 5:PRINT#1,"00000000011111111112222222
2233":PRESET(11,170):PRINT#1,"123456789
0123456789012345678901":RETURN
5225 GOSUB 5220:PRESET(5,185):COLOR 10:
PRINT#1,"ACCT BALANCE ";VM$;YY;" <RET>Q
uit"
5230 FOR Z=1 TO DM:T1(Z)=T0(Z)/DV:IF T1
(Z)<0 THEN T1(Z)=0 ELSE NEXT
5240 PRESET(10,0):DRAW"C13R245BD40L245B
D40R245BD40L245BD40R245"
5250 IF AA$="2" THEN 5300 ELSE C0=1:GR=
0:FOR Z=10 TO 255 STEP 8:C0=C0+1:GR=GR+
1:LINE(Z,160)-(Z+7,(160-T1(GR))),C0,BF:
IF GR=DM THEN 5270 ELSE IF C0>14 THEN C
0=1
5260 NEXT Z
5270 GOTO 5400
5300 CT=0:FOR Z=13 TO 255 STEP 8:CT=CT+
1:IF CT=DM THEN 5310 ELSE LINE(Z,(160-T
1(CT)))-(Z+8,(160-T1(CT+1))),15:NEXT Z
5310 '
5400 GOTO 5998
5500 PRINT:PRINT"* INPUT ACCEPTED. PLEA
SE WAIT":FOR Z=1 TO 31:D1(Z)=0:W1(Z)=0:
NEXT:FOR Z=1 TO DM:GOSUB 5000:IF FL=1 T
HEN D1(Z)=0 ELSE D1(Z)=VAL(MID$(TQ$(ZZ)
,41,16))
5505 NEXT Z
5510 FOR Z=1 TO DM:IF 0>(D1(Z)) THEN W1
(Z)=ABS(D1(Z)):D1(Z)=0
5520 NEXT Z
5525 IF PX=1 THEN 6400
5530 FOR Z=1 TO DM:D1(Z)=(D1(Z)/DV):W1(
Z)=(W1(Z)/DV):NEXT Z
5540 GOSUB 5220:PRESET(10,0):DRAW"C10R2
45BD40L245BD40R245BD40L245BD40R245":IF
AA$="2" THEN 5600 ELSE C0=1:GR=0:FOR Z=
10 TO 255 STEP 8:C0=C0+1:GR=GR+1
5550 IF ZX$="2" THEN LINE(Z,160)-(Z+7,(
160-D1(GR))),C0,BF:IF GR=DM THEN 5700 E
LSE IF C0>14 THEN C0=1
5560 IF ZX$="3" THEN LINE(Z,160)-(Z+7,(
160-W1(GR))),C0,BF:IF GR=DM THEN 5700 E
LSE IF C0>14 THEN C0=1
5570 NEXT Z:GOTO 5700
```

```
5600 CT=0:FOR Z=13 TO 255 STEP 8:CT=CT+
1:IF CT=DM THEN 5700
5610 IF ZX$="2" THEN LINE(Z,(160-D1(CT)
))-(Z+8,(160-D1(CT+1))),15
5620 IF ZX$="3" THEN LINE(Z,(160-W1(CT)
))-(Z+8,(160-W1(CT+1))),15
5630 NEXT Z:GOTO 5700
5700 PRESET(5,185):COLOR 4:IF ZX$="2" T
HEN PRINT#1,"DEPOSITS: ";VM$;YY;"    <RE
T> Quit" ELSE IF ZX$="3" THEN PRINT#1,"
WITHDRAWALS:";VM$;YY;" <RET> Quit"
5998 IF INKEY$<>"" THEN 5998
5999 A$=INKEY$:IF A$="" THEN 5999 ELSE
IF A$=CHR$(13) THEN SCREEN 0:COLOR 15,1
:GOTO 2000 ELSE 5999
6000 ' * PRINTER ROUTINES!! *
6010 CLS:PRINT"**** GRAPHICS MODE     (X
-Y PLOTTER) ****":PX=1:GOTO 5110
6020 CLOSE:IF SC=1 THEN DV=2 ELSE IF SC
=2 THEN DV=20 ELSE IF SC=3 THEN DV=200
ELSE IF SC=4 THEN DV=2000
6025 IF DV=2 THEN SX=1000 ELSE IF DV=20
 THEN SX=10000 ELSE IF DV=200 THEN SX=1
00000! ELSE IF DV=2000 THEN SX=1000000#
6030 GOTO 5205
6040 FOR Z=1 TO DM:T1(Z)=T0(Z)/DV:IF T1
(Z)<0 THEN T1(Z)=0 ELSE NEXT
6050 FOR Z=1 TO DM:GR(Z)=T1(Z):NEXT:NM$
="ACCOUNT BALANCE ":GOTO 8000
6400 IF ZX$="3" THEN 6500
6410 FOR Z=1 TO DM:GR(Z)=(D1(Z)/DV):NEX
T:NM$="DEPOSITS":GOTO 8000
6500 FOR Z=1 TO DM:GR(Z)=(W1(Z)/DV):NEX
T:NM$="WITHDRAWALS":GOTO 8000
6700 CLS:PRINT:PRINT"***** PRINTOUT OF
ALL TRANSACTIONS *****":PRINT"TO ABORT,
 PRESS <CTRL><STOP>..."
6705 LPRINT"     ***** HOME BANKING MANA
GER *****":LPRINT" [C]1986 ROGER STOCKB
URGER MSX SOFTWARE":LPRINT
6710 CLOSE:ON STOP GOSUB 3760:GOSUB 190
0:OPEN FI$ AS #1:FIELD #1,80 AS TS$:FOR
 Z=1 TO RT:GET #1,Z:LPRINT MID$(TS$,73,
2);"/";MID$(TS$,75,2);"/";RIGHT$(TS$,4)
;SPC(3);"Transact $";MID$(TS$,41,16)
6720 LPRINT SPC(13);"Balance  $";MID$(T
S$,57,16):LPRINT LEFT$(TS$,40):LPRINT:N
EXT Z:LPRINT:LPRINT"#### END OF REPORT
####":CLOSE:LPRINT CHR$(12):GOTO 2000
6800 CLS:LOCATE 0,10:PRINT"--------
 A B O R T E D     --------":PRINT:PRIN
T"Press any Key
6999 END
7000 ' * NO FILE-NAME ERROR *
7010 CLS:PRINT"/-----------------------
---------------\|  ACCESS TO THIS PART
OF THE PROGRAM  ||  IS NOT POSSIBLE UNT
IL OPTION #3 IS  ||  EXECUTED IN THE MA
IN MENU.          |";
7020 PRINT"\---------------------------
-----------/"
7030 PRINT:PRINT:PRINT"PRESS <RETURN> T
O RETURN TO MAIN MENU..."
7040 DEFUSR1=&H9F:X=USR1(1):GOTO 100
7990 CLEAR:RUN
8000 ' PRINTER ROUTINE - GR(X)=COMMON
8010 CLS:PRINT"***** PRINTER ROUTINES X
-Y PLOTTER *****":PRINT:PRINT"INSERT A4
 SHEET PAPER INTO PLOTTER,  ANDADJUST P
APER FEED, SO THAT THE TOP OF ITIS APPR
OX.  4mm UPWARDS FROM THE POINTERON THE
 PRINTHEAD.":PRINT
8012 PRINT"*** PRESS RETURN WHEN YOU AR
E READY: ***"
8015 A$=INKEY$:IF A$<>CHR$(13) THEN 801
5
8050 LPRINT CHR$(&H1B)+"#":LPRINT"R60,-
120":LPRINT"I":LPRINT"S1"
8060 LPRINT"Q1":LPRINT"C0":FOR A=1 TO D
M:LPRINT"R0,-8":LPRINT "P";USING"##";A:
LPRINT"R0,-8":NEXT:LPRINT"H":LPRINT"R25
,0"
8070 C=0:FOR A=1 TO DM:C=C+1:LPRINT"C";
C:LPRINT"R0,-1":LPRINT"J";GR(A);",0":LP
RINT"J0,-37":LPRINT"J";-GR(A);",0":LPRI
NT"J2,-2,";GR(A);",0,0,37,-2,2":LPRINT"
R";-GR(A);",-39":IF C=3 THEN C=0
8080 NEXT A
8090 LPRINT"C0":FOR A=1 TO DM*2:LPRINT"
J0,20":NEXT
8100 LPRINT"S2":LPRINT"R-60,-100":LPRIN
T"P";NM$;" FOR ";VM$;YY:LPRINT"S1"
8110 LPRINT"H":LPRINT"R25,10":LPRINT"J5
00,0"
8120 LPRINT"C1":FOR A=500 TO 0 STEP-250
:FOR Z=1 TO DM:LPRINT"J0,-40":NEXT Z:LP
RINT"J0,-20":LPRINT"R-125,0":FOR Z=1 TO
 DM*2:LPRINT"J0,20":NEXT Z:LPRINT"J0,20
":NEXT A:LPRINT"H":LPRINT"R25,10":LPRIN
T"R500,0"
8130 LPRINT"C0":FOR A=1 TO 6:LPRINT"R0,
17":NEXT A:LPRINT"P$";USING"########";SX
:LPRINT"R-125,96":LPRINT"P$";USING"####
###";(SX*.75):LPRINT"R-125,96":LPRINT"P
$";USING"########";(SX/2):LPRINT"R-125,9
6"
8140 LPRINT"P$";USING"########";SX*.25:L
PRINT"R-125,96":LPRINT"P$       0"
8150 LPRINT"R770,-100":LPRINT"S5":LPRIN
T"I":LPRINT"C1":LPRINT"P  HOME BANKING
MANAGER":LPRINT"HC3":LPRINT"R-200,-30":
LPRINT"J80,-40,-15,25,80,-40,-136,50":L
PRINT"R45,-80":LPRINT"S2":LPRINT"C0":LP
RINT"P[C]1986 ROGER STOCKBURGER MSX SOF
TWARE"
8160 LPRINT"C2,R200,-300,0,-70":LPRINT"
R-908,0":LPRINT"J0,500,0,500,0,356":LPR
INT"R906,0":LPRINT"J0,-500,0,-500,0,-35
7":LPRINT"H":LPRINT"R0,500,0,500,0,500"
:BEEP:LPRINT"C0,A"
8170 FOR A=1 TO 31:GR(A)=0:NEXT:PX=0:GO
TO 2000
9000 ' * SET BANK NAME *
9010 CLS:PRINT"/----------- SET BANK NA
ME: ----------\|";TAB(39);"|"
9020 PRINT:PRINT"PLEASE ENTER APPREVIAT
ED NAME OF BANK   YOU WISH TO OPERATE O
N:":PRINT"(=== MAXIMUM 5 LETTERS ===)":
PRINT:INPUT N$:N$=LEFT$(N$,5)
9030 FI$="A:"+N$+".DAT":FC$="A:"+N$+".C
OD"
9040 CLOSE:ON ERROR GOTO 9100
9050 OPEN FI$ FOR APPEND AS #1
```

```
9060 OPEN FC$ FOR APPEND AS #2
9070 PI$="PASS":GOTO 100
9100 ' * DISK ERROR *
9105 IF ERR=19 THEN RESUME 100
9110 PI$="":CLS:PRINT"========  D I S K
    E R R O R  ========"
9120 PRINT:IF ERR=70 THEN PRINT"= DISK
OFFLINE =" ELSE IF ERR=68 THEN PRINT"=
DISK WRITE PROTECTED =" ELSE IF ERR=66
THEN PRINT"= DISK FULL, START NEW DISK
=" ELSE IF ERR=53 THEN PRINT"= FILE NOT
 FOUND =":GOTO 9200
9130 IF ERR=69 THEN PRINT"= DISK I/O ER
ROR =":RUN
9140 PRINT:PRINT"--- PLEASE PRESS ANY K
EY TO CONTINUE ---"
9150 DEFUSR1=&H9F:X=USR1(1):RESUME 100
9200 '
9210 CLS:PRINT"====  F I L E    N O T
 F O U N D  ===="
9220 PRINT:PRINT"  The abbreviation of
the Bank you have  selected does not ex
ist on this disk."
9230 PRINT:PRINT"  Do you  wish  to est
ablish a  file on  the bank you have se
lected?":PRINT
9240 PRINT:PRINT"  [Y] - Yes,  [N] - No
, goto main menu"
9250 A$=INKEY$:IF A$="Y" OR A$="y" THEN
 9300 ELSE IF A$="N" OR A$="n" THEN RES
UME 100 ELSE 9250
9300 ' OPEN NEW FILE
9310 ON ERROR GOTO 9100
9315 CLOSE
9320 OPEN FI$ AS #1
9330 OPEN FC$ FOR OUTPUT AS #2
9340 CLOSE:PI$="PASS"
9350 RESUME 100
```

THIS PROGRAM IS MADE TO SUIT
THE SV-328 OR 318 EXPANDED.

```
10 ' Brian Parker 28/1/85
20 ' ********************
30 CLEAR100,53605!
40 ' copy rom to ram
50 DEFUSR0=53600!
60 FORX=53600!TO53678!:READA$:POKEX,VAL
("&h"+A$):NEXT
70 A=USR(0)
80 DATAf3,21,00,00,11,a0,8c,01,02,40,ed
,b0,3e,0f,d3,88,3e,fd,d3,8c,00,00,21,a0
,8c,11,00,00,01,02,40,ed,b0,3e,0f,d3,88
,3e,ff,d3,8c,21,00,40,11,a0,8c,01,02,40
,ed,b0,3e,0f,d3,88,3e,dd,d3,8c,21,a0,8c
,11,00,40,01,02,40,ed,b0,3e,dd,32,61,7b
,c9,00,00
90 ' changes to basic
100 ' creates two new commands
110 ' DUMP is a small screen dump
120 ' MAG is a magnified screen dump
130 ' N.B. these are at the expence of
  MDM and DIAL
140 POKE897,ASC("U")
150 POKE898,ASC("M")
160 POKE899,ASC("P")+128
170 POKE547,&H7F
180 POKE548,&HD1
190 POKE1123,ASC("A")
200 POKE1124,ASC("G")+128
210 POKE545,&H76
220 POKE546,&HD1
230 ' setting select key for a large
 dump and print key for small dump
240 FORX=&H7F98TO&H7FAB:READA$:POKEX,VA
L("&h"+A$):NEXT
250 DATAf5,79,fe,45,20,04,f1,c3,7f,d1,f
e,44,20,04,f1,c3,76,d1,f1,c9
260 POKE&HFF7D,&H7F
270 POKE&HFF7C,&H98
280 POKE&HFF7B,&HC3
290 'screen dump routine
300 'the list of colours to print
starts at 53607. Remember to fill
unused spaces with 255 so as not to
print unwanted colours.
305 REM*** REPLACE ANY OF FOLLOWING ff'
S WITH HEX No.OF DESIRED COLOR EG.WHITE
 = DECIMAL 15 = HEX 0F
310 FORX=53607!TO53930!:READA$:POKEX,VA
L("&h"+A$):NEXT
320 DATAff,ff,ff,ff,ff,ff,ff,ff,ff,ff,f
f,ff,ff,ff,ff:'colors to print
330 DATAf3,f5,c5,d5,e5,3e,1,18,07,f3,f5
,c5,d5,e5,3e,0,32,65,d1,3a,3a,fe,3d,c2,
a5,d2,3e,1,32,66,d1,1e,1b,cd,a7,d1,1e,4
1,cd,a7,d1,1e,08,cd,a7,d1,c3,d9,d1
340 DATAdb,12,e6,01,20,fa,7b,d3,10,af,d
3,11,3d,d3,11,c9,1e,0a,cd,a7,d1,1e,1b,c
d,a7,d1,1e,4b,cd,a7,d1,1e,00,cd,a7,d1,1
e,01,3a,65,d1,3d,20,2,1e,2,cd,a7,d1,c9
350 DATAdb,85,21,f7,ff,11,00,80,01,08,0
0,09,7d,fe,ff,20,2b,db,9a,e6,f0,f6,06,d
3,96,db,99,e6,22,ca,a5,d2,3a,65,d1,3d,2
0,13,3a,66,d1,3d,28,08,25,3e,1,32,66,d1
,18,05,3e,0,32,66,d1,cd,b7,d1,01,08,00,
09,01,81,08,ed,69,ed,61,00,00,00,db,84,
a2,37,20,1,3f
360 DATAcb,ec,ed,69,ed,61,0,0,0,db,84,3
0,0c,e6,f0,cb,3f,cb,3f,cb,3f,cb,3f,18,0
2,e6,0f,d9,06,0f,21,67,d1,4e,b9,28,07,2
3,10,f9,37,3f,18,01,37,d9,cb,ac
370 DATACB,1B,2B,10,bd,3a,65,d1,3d,20,2
0,3a,66,d1,3d,20,08,cb,23,cb,23,cb,23,c
b,23,06,4,cb,2b,cb,23,cb,11,cb,23,cb,1!
,10,f4,59,cd,a7,d1,CD,a7,d1,CB,3A,20,89
,7C,FE,17,28,3,c3,de,d1,7d,fe,f7,28,3,c
3,de,d1,3a,66,d1,3d,28,03,c3,de,d1,e1,D
1,C1,F1,FB,C9
```

?????????????????????????????? IDEAS ??????????????????????????????

As you know by now this issue contains the last edition of the Basic Tutorial, so for the next addition we must be able to fill those pages as to hopefully keep the magazine to this size if not bigger.

So we require help from you in the way of input.

It can be a letter of inquiry, a small cartoon, Ideas on a new column, send in your high game score or something like a blunder column for eg- I made a program for my kids which was fairly simple it was a program that asked them to add, multiply, subtract, divide and type in the correct answer, if they got it right it played a nice tune but if they got it wrong it played a dull tune, anyway after just on 2 hours making this program and testing it I starting getting a bit tired so I swiched off my computer to go out to watch football on the television. I no sooner got up from my chair when I realised what I had done (not saved my program) so after really telling myself off for being such an egg I then watched TV.

I'm sure there are a few "eggs" out there besides me, so come on we would like to hear from you.

Also we thought we would run a small competition this issue for ideas for the design of our front cover we leave it up to you, but please remember we are only allowed 2 colours or colors which ever you prefer.

We will send $10.00 to every winning entry that we will use on the front cover, so send them in.

Just a coloured sketch of what you think of, is fine by us we do not judge on neatness. Plus keep your ideas original.

Please remember that this magazine only exists because of the continued support by you, so it is really your magazine.

So please dont just sit there reading this and thinking 'some body else will send something in' because if we all thought like that the magazine would only consist of a front and back cover.

WE WOULD LIKE TO APOLOGIZE FOR THIS ISSUE BEING LATE BUT I HOPE YOU CAN UNDERSTAND THAT WHEN YOU DO SOMETHING THAT YOU ARE NOT USED TO IT IS VERY HARD.

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